

SGA 3, Exposé XII
Maximal Tori, the Weyl Group, Cartan Subgroups, and the
Reductive Center
English Reconstruction

A. Grothendieck

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From the present Exposé onwards, in contrast to the preceding Exposés, we shall use the known results on the structure of smooth algebraic groups over an algebraically closed field k , and above all Borel's theory of affine algebraic groups, expounded in the 1956/57 Chevalley Seminar, *Classification of Algebraic Lie Groups*.⁰ Following the usage in the theory of algebraic groups, we shall refer to that seminar by the siglum *BIBLE*. In the following Exposés we shall need, in particular, the results of *BIBLE* 4, 5, 6, and 7 (the number following *BIBLE* is the number of the Exposé). Moreover, it appears that scheme theory brings no notable simplification to Borel's theory as it is presented in *BIBLE*. For this reason it did not seem useful to reproduce that theory in the present Seminar: our purpose here is to deduce from results known over algebraically closed fields analogous results valid over an arbitrary base prescheme. (The situation will be different for Chevalley's structure theory of semisimple groups, which, it appears, is more advantageously treated *ab ovo* over an arbitrary base prescheme.)

In the oral Exposés (which we followed in Nos. 1 to 4), we restricted ourselves to affine group preschemes over S , relying essentially on the representability theorems of Exposé XI, No. 4. In the present notes (cf. Nos. 6 to 8), we show quickly how the affineness hypotheses can be eliminated by a simpler method that does not use the most delicate results of Exposé XI. For further generalizations of the results contained in the present Exposé, see also Exposés XV and XVI. (It is clear that the contents of Exposés XI, XII, XV, and XVI ought to be completely rewritten.)

1 Maximal tori

1.0. Let first G be an algebraic group over an algebraically closed field k . A *maximal torus* of G is an algebraic subgroup T of G which is a torus (here, since k is algebraically closed, this means that it is isomorphic to a group of the form $\mathbf{G}_m^r$) and which is maximal for this property. Notice that, since k is perfect, G_{red} is a subgroup of G , smooth over k , and hence is essentially an algebraic group in the sense of *BIBLE*. Moreover, every reduced subgroup of G (such as a torus) is automatically a subgroup of G_{red} . Consequently the maximal tori of G are precisely the maximal tori of G_{red} . When G is affine, and therefore G_{red} is affine, a fundamental theorem of Borel says that any two maximal tori of G are conjugate by an element of

$$G(k) = G_{\text{red}}(k)$$

(*BIBLE* 6, Theorem 4(c)); in particular, they have the same dimension. We call the common dimension of the maximal tori of G the *reductive rank* of G . The assumption that G be affine is in fact unnecessary. This follows from a theorem of Chevalley asserting that every connected smooth algebraic group over k is an extension of an abelian variety by an affine group; cf. No. 6. In Nos. 1 to 4 we shall nevertheless most often restrict ourselves to affine group preschemes over the base.

Let now G be a smooth algebraic group over k , and let T be a maximal torus of G . The centralizer $C(T)$ of T in G will be called the *Cartan subgroup of G associated with T* . For us it is a group subscheme defined by VIII 6.7. Since G is smooth over k , however, its Cartan subgroup C is also smooth over k , by XI 5.3. Thus C is in this case the unique group subscheme of G , smooth over k (that is, an algebraic subgroup of G in the sense of *BIBLE*), for which $C(k)$ is the subgroup of $G(k)$ centralizing $T(k)$; in other words, it is essentially the centralizer in the sense of *BIBLE*.

⁻¹Version xy of 2 December 2008.

⁰Editor's note: this seminar has since been published, in a form revised by P. Cartier, as volume 3 of Chevalley's collected works (Springer, 2005).

By the conjugacy theorem just cited, the Cartan subgroups associated with the various maximal tori are conjugate to one another and therefore have the same dimension. We call their common dimension the *nilpotent rank of G* ; it is equal to that of G_{red} . If $\rho_r(G)$ and $\rho_n(G)$ denote the reductive and nilpotent ranks of G , then

$$\rho_r(G) \leq \rho_n(G),$$

and the difference

$$\rho_u(G) = \rho_n(G) - \rho_r(G) = \dim(C/T)$$

might be called, when G is affine, the *unipotent rank of G* . If G is smooth, affine, and connected, then C is a connected nilpotent algebraic group (*BIBLE* 6, Theorem 6(a),(c)). Hence, by *BIBLE* 6, Theorem 2, it is isomorphic to the product

$$C_s \times C_u,$$

where $C_s = T$ is the original maximal torus (and *a fortiori* a maximal torus of C), while C_u is a smooth unipotent subgroup, that is, a successive extension of groups isomorphic to $\mathbf{G}_a$ (*BIBLE* 6, Theorem 1, Corollary 1, and *BIBLE* 7, Theorem 4). In this case one also has

$$\rho_u(G) = \dim C_u.$$

Remark 1.1. Besides the three notions of rank just specified for an affine algebraic group, two others are useful. The first is the *semisimple rank* $\rho_s(G)$, defined as the reductive rank of the quotient G/R , where R is the radical of G . The second is the *infinitesimal rank* $\rho_i(G)$, defined as the nilpotent rank of the Lie algebra of G , which will be defined and studied in the following Exposé. We shall not use them in the present Exposé. We record only the inequalities

$$\rho_s \leq \rho_r \leq \rho_n \leq \rho_i.$$

Lemma 1.2. *Let G be an algebraic group over the algebraically closed field k , let T be an algebraic subgroup of G , and let k' be an algebraically closed extension of k . Let G' and T' be the groups obtained from G and T by base extension. Then T is a maximal torus of G if and only if T' is a maximal torus of G' .*

This is an immediate consequence of the “principle of finite extension” (EGA IV, 9.1.1).

Definition 1.3. Let S be a prescheme, let G be an S -group prescheme of finite type, and let T be a group subscheme of G . One says that T is a *maximal torus of G* if:

- a) T is a torus (IX 1.3); and
- b) for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Remarks 1.4. It follows from 1.2 that, when S is the spectrum of an algebraically closed field, this recovers the usual definition, and that property 1.3 is stable under arbitrary base change. Notice that, by X 8.8, condition (a) in Definition 1.3 may be replaced by:

- a') T is of finite presentation and flat over S .

One should take care: a maximal torus in the sense of Definition 1.3 is indeed maximal among the subtori of G , as follows immediately from IX 2.9, but the converse cannot hold—say when the base S is connected—unless G actually admits a maximal torus in the sense of 1.3. This is not true in general, even when $S = \operatorname{Spec}(\mathbb{Z})$ and G is “semisimple”; see also 1.6. We shall see in Exposé XIV, however, that the converse is true when S is artinian, or when S is a local scheme and G is “reductive”: in that case every torus of G is contained in a maximal torus.

Definition 1.5. Let G be an algebraic group over a field k . The *reductive rank* (respectively the *nilpotent rank*, the *unipotent rank*, and so on) of G is the reductive rank (respectively, and so on) of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k .

By 1.2 and the compatibility of the formation of $\operatorname{Centr}_G(T)$ with base extension, the notions of rank introduced in 1.5 are invariant under extension of the base field. For algebraically closed k , they coincide with the notions introduced at the beginning of the present number.

Remark 1.6. It is not difficult to construct a smooth affine group scheme G over the spectrum S of a discrete valuation ring whose generic fiber is isomorphic to $\mathbf{G}_m$ and whose special fiber is isomorphic to $\mathbf{G}_a$. Such a G contains no torus other than the trivial torus T , reduced to the identity subgroup, and T is plainly not a maximal torus. More precisely, in the special fiber $G_0 = \mathbf{G}_{a,k}$, the torus T_0 is indeed maximal, whereas in the generic fiber $G_1 = \mathbf{G}_{m,K}$, the torus T_1 is no longer maximal; here k is the residue field and K the fraction field of the valuation ring. This example also shows that the reductive rank of G_s , for $s \in S$, is not a continuous function of s . Nevertheless, one has the following results.

Theorem 1.7. Let S be a prescheme and let G be a smooth affine S -group prescheme. For every $s \in S$, set

$$\rho_r(s) = \rho_r(G_s), \quad \rho_n(s) = \rho_n(G_s),$$

the reductive and nilpotent ranks of G_s as in 1.5. Then:

- a) The function ρ_r on S is lower semicontinuous, while ρ_n is upper semicontinuous. Hence $\rho_u = \rho_n - \rho_r$ is upper semicontinuous.
- b) The following conditions, each stable under arbitrary base change, are equivalent:
 - (i) the function ρ_r on S is locally constant;
 - (ii) G admits a maximal torus locally for the étale topology;
 - (ii bis) G admits a maximal torus locally for the faithfully flat quasi-compact topology.
- c) Let T_1, T_2 be two maximal tori of G , which in particular places us under the conditions of (b). Then T_1 and T_2 are locally conjugate for the étale topology: there exists a surjective étale morphism $S' \rightarrow S$ such that the subgroups $(T_1)_{S'}$ and $(T_2)_{S'}$ of $G_{S'}$ are conjugate by a section of $G_{S'}$ over S' .
- d) Under the conditions of (b), that is, when ρ_r is locally constant, ρ_n is locally constant as well, and hence so is $\rho_u = \rho_n - \rho_r$.

Proof. a) Observe that for every morphism $S' \rightarrow S$, if $G' = G \times_S S'$, then the functions ρ'_r , and so on, on S' , defined in terms of G' as ρ_r , and so on, are defined in terms of G , are obtained simply by composing the latter functions with $S' \rightarrow S$. When $S' \rightarrow S$ is faithfully

flat and quasi-compact, it follows that ρ' is continuous, respectively upper semicontinuous, respectively lower semicontinuous, if and only if ρ is, since the Zariski topology on S is the quotient of that on S' (SGA 1, VIII 4.3). Consequently, the assertions in (a) are local for the faithfully flat quasi-compact topology.

Let $s \in S$. We wish to show that the set U of $t \in S$ such that

$$\rho_r(t) \geq \rho_r(s) \quad (\text{respectively } \rho_n(t) \leq \rho_n(s))$$

is a neighborhood of s . By the principle of finite extension, there exists a finite extension k of $\kappa(s)$ such that G_k admits a maximal torus. There then exist an open neighborhood U of s and a finite, surjective, locally free morphism $S' \rightarrow U$ such that the fiber S'_s is $\kappa(s)$ -isomorphic to $\text{Spec}(k)$ (cf. EGA III, 10.3.2, where the noetherian hypothesis is plainly unnecessary). Since $S' \rightarrow S$ is an open morphism (SGA 1, IV 6.6), we are reduced to the case $S' = S$, that is, to the case in which G_s admits a maximal torus T_s . Moreover, by XI 5.8(a), after possibly replacing S once more by an étale S' over S , equipped with a point s' above s , we may suppose that T_s is the fiber at s of a subtorus T of G . Then, for every $t \in S$,

$$\rho_r(t) = \rho_r(G_t) \geq \dim T_t = \dim T_s = \rho_r(G_s) = \rho_r(s),$$

which proves that ρ_r is lower semicontinuous.

On the other hand, by XI 5.3, the functor

$$\text{Centr}_G(T)$$

is represented by

$$C = \text{Centr}_G(T),^1$$

a closed group subscheme of G , smooth over S . There is therefore a neighborhood U of s such that $t \in U$ implies

$$\dim C_t = \dim C_s = \rho_n(s).$$

The upper semicontinuity of ρ_n now follows from the relation

$$\rho_n(t) \leq \dim C_t \quad \text{for every } t \in S,$$

which is itself contained in the following purely geometric lemma.

Lemma 1.8. *Let k be a field, let G be a smooth affine algebraic group over k , let T be a torus in G , and let C be its centralizer. Then*

$$\rho_n(G) \leq \dim C.$$

Indeed, we may suppose k algebraically closed, so that T is contained in a maximal torus T' . If C' is the centralizer of T' , then $C' \subset C$, and hence $\dim C' \leq \dim C$. This proves the lemma.

b) If ρ_r is locally constant, then for every torus T in G and every $s \in S$, if T_s is a maximal torus in G_s , there exists an open neighborhood U of s such that $T|_U$ is a maximal torus in $G|_U$. Applying the argument of (a), we see that (i) implies (ii bis). Conversely, (ii bis) implies (i), because when G admits a maximal torus T , it is clear that

$$\rho_r(s) = \dim T_s$$

¹Editor's note: the text has been modified here to introduce the prescheme $C = \text{Centr}_G(T)$ representing the functor $\text{Centr}_G(T)$.

is a locally constant function of s ; and we observed at the beginning of the proof of (a) that continuity of ρ_r is local for the faithfully flat topology. Thus (i) is equivalent to (ii bis), while (ii) plainly implies (ii bis); it remains to prove the converse implication (i) $\Rightarrow$ (ii).

For this purpose, introduce the functor F of XI 4.1,² which is a smooth separated prescheme over S , and consider the subfunctor $\mathcal{T}$ of F whose value on an S' over S is the set of maximal tori in $G_{S'}$. By the preceding observation that, under (i), a torus in $G_{S'}$ which is maximal in the fiber over a point $s \in S'$ is maximal over an open neighborhood of s , the functor $\mathcal{T}$ is represented by an open subprescheme of F , and is therefore smooth and separated over S . Since the structural morphism $\mathcal{T} \rightarrow S$ is plainly surjective, it admits a section locally for the étale topology by XI 1.10. This proves (i) $\Rightarrow$ (ii).

c) This is an immediate consequence of XI 5.4 bis, taking account of Borel's conjugacy theorem recalled at the beginning of this number.

d) By the observation at the beginning of the proof of (a), and in view of (b), we may suppose that G admits a maximal torus T . If C is its centralizer, then C is representable and smooth over S by XI 5.3, so the function

$$s \longmapsto \rho_n(s) = \dim C_s$$

is indeed locally constant.

This completes the proof of 1.7. We shall refer to the conditions in 1.7(b) by saying that, in this case, G is of *locally constant reductive rank*. We record the following consequence.

Corollary 1.9. *Let G be as in 1.7, and let $s \in S$ be such that $\rho_u(s) = 0$, that is, $\rho_r(s) = \rho_n(s)$ (equivalently, the Cartan subgroups of $G_{\bar{k}}$ are tori, where $\bar{k}$ is an algebraic closure of $\kappa(s)$). Then there exists an open neighborhood U of s on which ρ_r and ρ_n are constant. In particular, for every $t \in U$, the unipotent rank $\rho_u(t)$ of G_t is zero.*

Indeed, this follows immediately from 1.7(a) and the inequality $\rho_r(t) \leq \rho_n(t)$ for every $t \in S$.

Notice also that, in the course of proving (b), we proved the following result.

Corollary 1.10. *Let G be as in 1.7, and suppose that G has locally constant reductive rank. Consider the functor*

$$\mathcal{T} : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens})$$

such that, for every S' over S ,

$$\mathcal{T}(S') = \{\text{maximal tori of } G_{S'}\}.$$

Then $\mathcal{T}$ is represented by a smooth, separated prescheme of finite type over S .

It remains only to verify that $\mathcal{T}$ is of finite type over S . Since the question is local for the faithfully flat quasi-compact topology, we may suppose that G admits a maximal torus T . By XI 5.3 bis and the bis version of XI 5.5, $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$ are represented by preschemes $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$, and $\mathcal{T}$ is isomorphic to

$$G/\text{Norm}_G(T).^3$$

The morphism $G \rightarrow \mathcal{T}$ defined by $g \mapsto \text{int}(g)(T)$ is surjective. Since G is quasi-compact over S , so is $\mathcal{T}$, which completes the proof.

²Editor's note: this is the functor of multiplicative-type subgroups of G .

³Editor's note: the text has been modified here to introduce the preschemes $\text{Norm}_G(T)$ and $G/\text{Norm}_G(T)$.

Remarks 1.11. a) The prescheme $\mathcal{T}$ of 1.10 will properly be called the *prescheme of maximal tori* of G . We shall see in No. 5 that it is in fact affine over S . This can be checked directly when G is isomorphic to a closed group subprescheme of a prescheme of the form $\mathrm{GL}(n)$, using XI 4.6 and observing that, by the argument in the proof of 1.7(b), $\mathcal{T}$ identifies with a subprescheme both open and closed in the prescheme F representing the multiplicative-type subgroups of G .

b) It is possible to prove 1.7, and hence 1.9, without using the delicate results of Exposé XI, using only the elementary results XI 3.12 and XI 6.2 and working solely with multiplicative-type groups finite over the base (morally, the groups ${}_nT$, where T is a maximal torus); compare No. 7. The same observation applies to the proof of 1.10.

We end this number with examples in which there is a unique maximal torus.

Proposition 1.12. *Let G be an S -group prescheme of multiplicative type and finite type. Then G admits a unique maximal torus, and every torus in G is contained in this maximal torus.*

Uniqueness plainly follows from the latter assertion, which characterizes the maximal torus as the largest subtorus of G . Uniqueness also makes existence a question local for the faithfully flat quasi-compact topology. We may therefore suppose that G is diagonalizable, say $G = D_S(M)$, where M is a finitely generated commutative group. Let M_0 be the quotient of M by its torsion subgroup. We claim that the torus

$$T = D_S(M_0)$$

in G is a maximal torus and the largest subtorus. Indeed, a subtorus T' of G is locally diagonalizable for the fpqc topology. Thus, in order to prove $T' \subset T$, we may suppose that T' is diagonalizable, and hence of the form $D_S(N)$, where N is a free quotient of M (VIII 1.4 and 3.2(b)). It follows that N is a quotient of M_0 , and therefore that $T' \subset T$. Since the construction of $T = D_S(M_0)$ is compatible with arbitrary base extension, this proves at the same time that T is a maximal torus of G , and completes the proof. When G is smooth over S , Proposition 1.12 has the following generalization.

Proposition 1.13. *Let G be an S -group prescheme of finite presentation over S . Suppose that, locally for the faithfully flat quasi-compact topology, G admits a central maximal torus. Then G admits globally a unique maximal torus,⁴ and it is the largest subtorus of G .*

Here again, uniqueness follows trivially from the last assertion and makes existence a local question for the fpqc topology. We may therefore suppose that G admits a central maximal torus T . We shall show that every torus R of G is contained in T . This follows from the next lemma.

Lemma 1.14. *Let G be an S -group prescheme of finite presentation over S , let T be a maximal torus of G , let R be a subtorus of G , and suppose that T and R commute. Then $R \subset T$.*

Indeed, since R and T commute, the morphism

$$R \times_S T \longrightarrow G, \quad (r, t) \longmapsto r \cdot t,$$

is a group homomorphism. By IX 6.8 it therefore has an image subgroup T' in G , which is a multiplicative-type group quotient of $R \times_S T$, hence a torus, and plainly contains T .

⁴Which is central!

Since T is a maximal torus, $T' = T$ by 1.4, and therefore $R \subset T$. This proves 1.14 and hence 1.13. In particular, using 1.7(b), we obtain:

Corollary 1.15. *Let G be a commutative, smooth, affine S -group prescheme of locally constant reductive rank. Then G admits a unique maximal torus, and this torus contains every subtorus of G .*

Corollary 1.16. *Let G be a smooth affine S -group prescheme. Suppose that, for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure $\bar{k}$ of $\kappa(s)$, the geometric fiber $G_{\bar{s}}$ is a connected nilpotent algebraic group (in the sense of BIBLE, that is, the group $G(\bar{k})$ of its $\bar{k}$ -valued points is nilpotent). Suppose moreover that G has locally constant reductive rank. Then G admits a unique maximal torus T ; moreover, T is central and is the largest subtorus of G .*

By 1.7(b), G admits a maximal torus locally for the fpqc topology, and by 1.13 we are reduced to proving that a maximal torus of G is central. By IX 5.6(b), we are reduced to the case in which S is the spectrum of a field, which we may suppose algebraically closed. Then $T(k)$ lies in the center of $G(k)$ by BIBLE 6, Theorem 2. This implies that T lies in the center of G , because $\text{Centr}_G(T)$ is a closed subscheme of G (VIII 6.6) containing all the points of $G(k)$, and is therefore equal to G by the Nullstellensatz, since G is reduced.

Finally, for later reference, we record the following elementary proposition, which we have already used implicitly.

Proposition 1.17. *Let $G \supset H \supset T$ be group preschemes of finite type over S . The following conditions are equivalent:*

- (i) T is a maximal torus of G ;
- (ii) T is a maximal torus of H , and for every $s \in S$ one has equality of reductive ranks

$$\rho_r(G_s) = \rho_r(H_s).$$

2 The Weyl group

Let first k be an algebraically closed field, and let G be a smooth affine algebraic group over k . If T is a maximal torus, C its centralizer, and N its normalizer, then by XI 5.9 these are smooth closed subgroups of G , and C is an open subgroup of N . Thus

$$W = N/C$$

is a finite étale group over k , and is consequently determined by the group $W(k)$ of its k -valued points:

$$W = W(k)_k$$

(the constant group defined by the ordinary finite group $W(k)$). The finite group $W(k)$ will be called the *geometric Weyl group*, or simply the *Weyl group* when no confusion is likely, of G relative to T . By Borel's conjugacy theorem, the Weyl groups relative to the various maximal tori are isomorphic to one another; this is why one sometimes speaks of "the" Weyl group of G , without specifying a maximal torus. Since the formation of C , N , and N/C commutes with arbitrary base extension, if k' is an algebraically closed extension of k , the geometric Weyl group of $G_{k'}$ relative to $T_{k'}$ is canonically isomorphic to that of

G relative to T . Consequently, “the” geometric Weyl group of G , properly speaking an isomorphism class of ordinary finite groups, coincides with that of $G_{k'}$.

This permits us, when G is a smooth affine algebraic group over an arbitrary field k , to speak of the *geometric Weyl group of G* as the isomorphism class of the geometric Weyl group of $G_{k'}$, where k' is any algebraically closed extension of k . When G admits a maximal torus T , we can form

$$C = \text{Centr}_G(T), \quad N = \text{Norm}_G(T), \quad W = N/C$$

as above. The latter is a finite étale group over k , called the *Weyl group of G relative to T* . The geometric Weyl group is then simply the class of the group of points of W with values in any algebraically closed extension k' of k . In general, knowledge of the geometric Weyl group $W(k')$ plainly no longer suffices to reconstruct the algebraic group W : one must also know the action on $W(k')$ of the Galois group of the separable algebraic closure of k in k' .

Finally, let G be a group prescheme over an arbitrary base S , smooth and affine over S , and let T be a maximal torus of G . By XI 5.9 we may again form the group

$$W(T) = N(T)/C(T),$$

which is an étale, separated, quasi-finite S -group prescheme. Its geometric fibers, relative to algebraic closures of the residue fields $\kappa(s)$, $s \in S$, are the geometric Weyl groups of the fibers G_s . Thus XI 5.10 gives information about the variation of these groups with $s \in S$. We can make this information more precise and generalize it as follows.

Theorem 2.1. *Let S be a prescheme and let G be a smooth affine S -group prescheme. For every $s \in S$, let $w(s)$ be the geometric Weyl group of G_s , hence an isomorphism class of finite groups. On the set E of such classes, introduce the following preorder: $w \leq w'$ if and only if w and w' can be represented by finite groups W and W' such that W is isomorphic to a quotient of a subgroup of W' . With this convention:*

- a) *The function $s \mapsto w(s)$ from S to E is lower semicontinuous.*
- b) *Suppose that G has locally constant reductive rank. Then the following conditions are equivalent:*
 - (i) *the function $s \mapsto w(s)$ is locally constant;*
 - (i bis) *the function $s \mapsto \text{card } w(s)$ is locally constant;*
 - (ii) *locally for the étale topology (or merely for the fpqc topology), there exists a maximal torus T such that $W(T)$ is finite over its base;*
 - (ii bis) *for every S' over S and every maximal torus T of $G_{S'}$, the associated Weyl group $W(T)$ is finite over S' .*

Proof. a) Arguing as in 1.7(a), in order to prove that for every $s \in S$ there exists an open neighborhood U of s such that $t \in U$ implies $w(t) \geq w(s)$, we are reduced to the case in which there is a torus R in G such that R_s is a maximal torus in G_s . Put

$$W(R) = \text{Norm}_G(R) / \text{Centr}_G(R).$$

As in XI 5.9, this is an étale, separated, quasi-finite group prescheme over S . For $t \in S$, let $w'(t) \in E$ be its geometric fiber at t . Since R_s is a maximal torus in G_s , and the

formation of Norm, Centr, and Norm / Centr commutes with arbitrary base extension, in particular with passage to fibers, one has

$$w(s) = w'(s).$$

We claim moreover that for t near s ,

$$w(t) \geq w'(t) \quad \text{and} \quad w'(t) \geq w'(s),$$

which will suffice to prove (a). These two inequalities are contained in the following two lemmas.

Lemma 2.2. *Let S be a prescheme, and let W be an étale, separated, quasi-finite S -group prescheme. For every $s \in S$, let $f(s)$ be the class of the geometric fiber of W at s , an element of the ordered set E of isomorphism classes of finite groups introduced in 2.1. Then the function $f : S \rightarrow E$ is lower semicontinuous. It is constant in a neighborhood of $s \in S$ if and only if the function $s \mapsto \text{card } f(s)$ is constant there, and this holds if and only if W is finite over S above some open neighborhood U of s .*

This result refines, in group-theoretic terms, the result used in the proof of XI 5.10. We give only a sketch of the proof, which is entirely standard. As usual, we are reduced to the case in which S is affine and noetherian. The function f is immediately seen to be constructible (EGA 0_{III}, 9.3.1 and 9.3.2, together with the sorites of EGA IV, 9), and EGA 0_{III}, 9.3.4 reduces semicontinuity to proving that if t is a generalization of s , then

$$f(t) \geq f(s).$$

By EGA II, 7.1.7, this reduces us to the case in which S is the spectrum of a discrete valuation ring, which may be supposed complete with algebraically closed residue field. Let s be the closed point of S . Since W is étale and separated over S , it contains a subscheme W' , both open and closed and finite over S , such that $W'_s = W_s$ (EGA II, 6.2.6); it is immediate here that W' is a subgroup of W . Moreover, since W' is finite étale over $S = \text{Spec}(V)$, with V complete and with algebraically closed residue field, it is a constant group, hence of the form A_S , where

$$A = W'(\kappa(s)) = W(\kappa(s))$$

has class $f(s)$. If B is the geometric fiber of W at the generic point t of S , there is therefore a canonical monomorphism $A \rightarrow B$, proving $f(s) \leq f(t)$. Notice that this argument actually proves semicontinuity for an order on E finer than the one specified in 2.1.

The equivalence between constancy of f at s and constancy of $s \mapsto \text{card } f(s)$ follows from the fact that, for $w, w' \in E$,

$$w \leq w' \quad \text{and} \quad \text{card } w = \text{card } w' \quad \implies \quad w = w'.$$

The equivalence of this condition with finiteness of W over a neighborhood of s is independent of the group structure on W ; it was noted after XI 5.10, and also follows readily from the preceding argument using the valuative criterion of properness (EGA II, 7.3.8).

Lemma 2.3. *Let G be a smooth affine algebraic group over an algebraically closed field k , and let $R \subset T$ be two subtori. Let $W(R)$ and $W(T)$ be the two associated finite groups of XI 5.9, obtained as the quotient of the normalizer by the centralizer. Then $W(R)$ is isomorphic to a quotient of a subgroup of $W(T)$.*

Indeed, consider the diagram

$$\begin{array}{ccccc}
 T & \hookrightarrow & C(T) & \hookrightarrow & C(R) \\
 & & \downarrow & & \downarrow \\
 & & N(T) \cap N(R) & \hookrightarrow & N(R) \\
 & & \downarrow & & \\
 & & N(T) & &
 \end{array}$$

Then $(N(T) \cap N(R))/C(T)$ is a subgroup of

$$W(T) = N(T)/C(T),$$

and there is an evident homomorphism

$$(N(T) \cap N(R))/C(T) \longrightarrow W(R) = N(R)/C(R).$$

Everything reduces to proving that this last homomorphism is surjective, that is, that for every k -valued point g of $N(R)$, there exists a k -valued point c of $C(R)$ such that cg normalizes T , or equivalently such that

$$\text{int}(c)(\text{int}(g)T) = T.$$

For this, it suffices to observe that $\text{int}(g)T$ is a torus of $N(R)$, hence of $C(R)$, which is an open subgroup of $N(R)$. The tori T and $\text{int}(g)T$ are maximal tori of $C(R)$, since they are maximal in G , and the assertion follows from Borel's conjugacy theorem.

This proves 2.3, and hence 2.1(a).

b) We have already observed that (i) and (i bis) are plainly equivalent. They imply (ii bis), by the converse part of 2.2 or, equivalently, XI 5.10. Condition (ii bis) implies (ii) by 1.7(b). Finally, (ii) implies (i bis): as in 1.7(a), condition (i) is local for the fpqc topology, so we may suppose that G admits a maximal torus T such that $W(T)$ is finite over S , and XI 5.10 again gives the conclusion. This completes the proof of 2.1.

3 Cartan subgroups

Definition 3.1. Let G be a smooth group prescheme of finite type over a prescheme S . A *Cartan subgroup* of G is a group subscheme C of G , smooth over S , such that for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $C_{\bar{s}}$ is a Cartan subgroup, in the sense of No. 1, of $G_{\bar{s}}$.

It is immediate that if C is a Cartan subgroup of G , then for every S' over S , $C_{S'}$ is a Cartan subgroup of $G_{S'}$. If S is the spectrum of an algebraically closed field, this recovers the notion recalled in No. 1. Finally, the property that a group subscheme C of G be a Cartan subgroup is plainly local for the faithfully flat quasi-compact topology.

Theorem 3.2. *Let G over S be as in 3.1, and suppose that G is affine over S and has locally constant reductive rank. Then the map*

$$T \longmapsto \text{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups⁵ of G . If C corresponds to T , then T is the unique maximal torus of C .

If T is a maximal torus of G , the functor $\text{Centr}_G(T)$ is represented, by XI 5.3 or 6.2, by a closed smooth subscheme

$$C = \text{Centr}_G(T)$$

of G , and the definitions show that C is a Cartan subgroup of G . Moreover, T is plainly a maximal torus of C , and, being central, it is the unique maximal torus of C by 1.13. Thus the map $T \mapsto \text{Centr}_G(T)$ from the set of maximal tori to the set of Cartan subgroups is injective; it remains to prove surjectivity.

Let C be a Cartan subgroup of G . We shall prove that $C = \text{Centr}_G(T)$ for a maximal torus T of G . It is enough to find a maximal torus T of C , for then T is a maximal torus of G : for every $s \in S$, the groups G_s and C_s have the same reductive rank. By IX 5.6(b), T lies in the center of C , so

$$C \subset C' = \text{Centr}_G(T).$$

The smooth subgroup C of the smooth group C' over S coincides with C' fiber by fiber; hence $C = C'$.

Since G has locally constant reductive rank, so does C . Therefore C admits a maximal torus locally for the étale topology by 1.7(b); that torus is central by the preceding argument, and 1.13 then shows that C itself admits a maximal torus. This completes the proof.

Corollary 3.3. *Let G be a smooth affine group prescheme over S , of locally constant reductive rank, and let*

$$\mathcal{C} : (\text{Sch})_S^\circ \longrightarrow (\text{Ens})$$

be the functor defined by

$$\mathcal{C}(S') = \{\text{Cartan subgroups of } G_{S'}\}.$$

Then $\mathcal{C}$ is isomorphic to the functor $\mathcal{T}$ of 1.10, and is therefore represented by the same smooth separated prescheme of finite type over S .

Corollary 3.4. *Under the hypotheses of 3.2, if $C = \text{Centr}_G(T)$, then*

$$\text{Norm}_G(C) = \text{Norm}_G(T).$$

Remark 3.5. When G does not have locally constant reductive rank, it may nevertheless admit Cartan subgroups. For example, if G has connected nilpotent fibers, then G is a Cartan subgroup of itself, although it need not have locally constant reductive rank; see 1.6. In Exposé XV we develop the theory of Cartan subgroups without assuming either that G is affine over S or that its reductive rank is locally constant, using the theory of regular elements from the next Exposé. See also Nos. 6 and 7 below for the removal of the affineness hypothesis.

⁵The proof given here in fact proves Theorem 3.2 only for closed Cartan subgroups of G . However, 7.1(a) establishes Theorem 3.2 as stated, and implies that the Cartan subgroups of G are closed.

4 The reductive center

Definition 4.1. Let S be a prescheme, let G be an S -group of finite presentation over S with affine fibers, and let Z be a group subscheme of G . We say that Z is a *reductive center* of G if:

- (i) Z is central and of multiplicative type;
- (ii) for every base change $S' \rightarrow S$ and every central homomorphism

$$u : H \longrightarrow G_{S'},$$

where H is a group of multiplicative type and of finite type over S' , the homomorphism u factors through $Z_{S'}$.

(For a variant of this notion when the fibers of G are not assumed affine, see 8.6.)

Such a Z is necessarily of finite type over S , since its fibers are, and is therefore uniquely determined as the largest central subgroup of multiplicative type in G . It is easy to give examples in which G , smooth and affine over S , admits a largest central subgroup Z of multiplicative type but Z is not a reductive center—that is, G admits no reductive center; see, for example, 1.6. It follows readily, however, from IX 6.8 that a subgroup Z of G is a reductive center if and only if it is a largest central subgroup of multiplicative type and retains this property after every base change. See also 4.3 below.

It is clear from 4.1 that if Z is the reductive center of G , then $Z_{S'}$ is the reductive center of $G_{S'}$ after every base change $S' \rightarrow S$. This fact and the uniqueness of the reductive center, together with faithfully flat quasi-compact descent (SGA 1, VIII), give:

Proposition 4.2. *Let G be an S -group prescheme of finite presentation over S , with affine fibers. If G admits a reductive center locally for the faithfully flat quasi-compact topology, then it admits a reductive center. For Z to be a reductive center of G , it is necessary and sufficient that it be so locally for the fpqc topology.*

We also record:

Proposition 4.3. *Let G be an S -group prescheme of finite presentation with affine fibers, and let Z be a group subscheme. For Z to be a reductive center of G , it is necessary and sufficient that it be of multiplicative type and that, for every $s \in S$, Z_s be a reductive center of G_s .*

Indeed, IX 5.6(b) first shows that Z is central. Since the conditions under consideration are stable under base change, it remains to prove that every central homomorphism

$$u : H \longrightarrow G,$$

with H of multiplicative type and of finite type, factors through Z . Since Z is central, u and the canonical immersion $v : Z \rightarrow G$ define a group homomorphism

$$w : H \times_S Z \longrightarrow G.$$

By IX 6.8, this homomorphism has a group image K , which is a subgroup of multiplicative type in G . Everything reduces to showing that $K = Z$. This holds fiber by fiber by the hypothesis on Z , and IX 5.1 bis now applies, completing the proof of 4.3.

In criterion 4.3, the hypothesis that Z_s be the reductive center of G_s for every $s \in S$ is in fact purely geometric: it is enough to verify it after passing to an algebraic closure of $\kappa(s)$, as follows from the second assertion of 4.2.

Theorem 4.4. *Let G be an affine algebraic group over a field k . Then G admits a reductive center.*

Since the center of G is represented by a closed subgroup of G by VIII 6.7, we are immediately reduced to the case in which G is commutative. Moreover, by 4.2 we may suppose that k is algebraically closed. We shall see in Exposé XVII that in this case G can be written as a product $Z \times U$, where Z is of multiplicative type and U is “unipotent”, from which it follows at once that Z is a reductive center of G . We give here a proof independent of the general structure theorem for commutative algebraic groups just indicated.

By IX 6.8, the set of subgroups H of multiplicative type in G is filtered upward. We show that it has a largest element.

When G is smooth over k , the structure theorem BIBLE 4, Th. 4 gives

$$G = G_s \times G_u,$$

where G_s is of multiplicative type and G_u is “unipotent”: here this means that G_u has a composition series whose factors are subgroups of $\mathbb{G}_a$, smooth if desired. Indeed, G_u/G_u^0 is a unipotent group by BIBLE 4, Corollary to Th. 3, hence a p -group, where p is the characteristic exponent, by BIBLE 4, Prop. 4. On the other hand, G_u^0 has a composition series with smooth connected one-dimensional quotients by BIBLE 6, Th. 1, Cor. 1, and these quotients are isomorphic to $\mathbb{G}_a$ by BIBLE 7, Th. 4. Every homomorphism from a group of multiplicative type to $\mathbb{G}_a$, and therefore also to G_u , is trivial, as follows from the lemma below. Thus every subgroup of multiplicative type in G is contained in G_s .

In the general case, consider the subgroup

$$G_0 = G_{\text{red}}$$

of G . It is smooth over k , and hence by the preceding case has a largest subgroup Z_0 of multiplicative type. The subgroups of multiplicative type in G containing Z_0 correspond to the subgroups of multiplicative type in $G' = G/Z_0$. We have an exact sequence

$$0 \longrightarrow G_0/Z_0 \longrightarrow G/Z_0 \longrightarrow G/G_0 \longrightarrow 0,$$

where G_0/Z_0 has no subgroup of multiplicative type other than the unit group. Since every subgroup of a group of multiplicative type is again of multiplicative type by IX 8, every subgroup H of multiplicative type in G/Z_0 satisfies

$$H \cap (G_0/Z_0) = 0;$$

thus $H \rightarrow G/G_0$ is injective. The algebraic group G/G_0 is radicial, hence finite over k , so H is itself radicial and its rank is bounded above by that of G/G_0 . Consequently the family of subgroups of multiplicative type in G has a largest element, say Z .

We claim that G/Z has no subgroup of multiplicative type other than the unit subgroup. Indeed, a commutative extension of two algebraic groups of multiplicative type is of multiplicative type (7.1.1). If Z' is the inverse image in G of a subgroup of multiplicative type in G/Z , then Z' is of multiplicative type. The maximality of Z gives $Z' = Z$, and hence $Z'/Z = 0$.

The “principle of finite extension” now readily shows that for every extension K of k ,

$$(G/Z)_K = G_K/Z_K$$

likewise has no subgroup of multiplicative type other than the unit group.

We can now prove that Z is a reductive center of G . Let

$$u : H \longrightarrow G_S$$

be a homomorphism of S -groups, where S is a prescheme over k and H is of multiplicative type and of finite type over S . We must show that u factors through Z_S , or equivalently that the composite

$$H \longrightarrow G_S \longrightarrow (G/Z)_S = G_S/Z_S$$

is trivial. Put $U = G/Z$. Every homomorphism $H \rightarrow U_S$ is trivial: by IX 5.2 one reduces to the case where S is the spectrum of a field K , and IX 6.8 then reduces the claim to the fact that U_K has no nontrivial subgroup of multiplicative type. This proves 4.4, modulo the following lemma.

Lemma 4.4.1. *Let H be an S -group prescheme of multiplicative type. Then every homomorphism of S -groups*

$$u : H \longrightarrow \mathbb{G}_a$$

is trivial.

Regard $E = \mathcal{O}_S^2$ as an extension with submodule $E' \simeq \mathcal{O}_S$ and quotient $E'' \simeq \mathcal{O}_S$. Then $\mathbb{G}_a$ identifies with the prescheme of automorphisms of this extension. A homomorphism $u : H \rightarrow \mathbb{G}_a$ therefore amounts to an H -module structure on E preserving the extension, with E' stable under H and with the induced actions on E' and E'' trivial. By I 4.7.3, H acts trivially on E ; hence u is trivial.

Remark 4.4.2. If G is a nonzero abelian variety over k , then G admits no reductive center in the sense of 4.1 with the restriction “ G affine” omitted. Indeed, if n is prime to the characteristic, then ${}_nG$ is étale over k , of order prime to the characteristic, and hence of multiplicative type. The family of the ${}_nG$ is schematically dense in G . A reductive center, if it existed, would therefore equal G , which is impossible because G is not of multiplicative type. This explains the requirement in 4.1 that G have affine fibers.

Lemma 4.5. *Let S be a prescheme, let G be a smooth affine S -group prescheme with connected fibers, let T be a maximal torus of G , and let $u : H \rightarrow G$ be a central homomorphism, where H is an S -group prescheme of multiplicative type and of finite type. Then u factors through T .*

Let $C = \text{Centr}_G(T)$. By XI 5.3 it is a closed smooth group subprescheme of G , hence affine over S . Since T is central in C , it is invariant, and the quotient group

$$U = C/T$$

is represented by VIII 5.1. Because u is central it factors through C , and it remains only to show that the composite

$$H \longrightarrow C \longrightarrow U = C/T$$

is trivial. By IX 5.2 we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. Then BIBLE 6, Th. 2 shows that U is a connected algebraic group, smooth over k , which is “unipotent”; as recalled above, it has a composition series with quotients isomorphic to $\mathbb{G}_a$. Every homomorphism from the finite-type group H of multiplicative type to U is therefore trivial. This proves 4.5.

Corollary 4.6. *Let G be as in 4.5. If G admits a reductive center, then that center is contained in every maximal torus of G .*

Theorem 4.7. *Let S be a prescheme and let G be an S -group prescheme, smooth and affine over S , with connected fibers.*

- (a) *For every $s \in S$, let $z(s)$ denote the type of the reductive center of G_s , which is defined by 4.4. Order the set E of isomorphism classes of finite-type $\mathbb{Z}$ -modules by declaring the class of M to be greater than that of M' when M' is isomorphic to a quotient of M . Then the map*

$$S \longrightarrow E, \quad s \longmapsto z(s),$$

is lower semicontinuous.

- (b) *The group G admits a reductive center Z if and only if the preceding function $z : S \rightarrow E$ is locally constant. If this holds, then G/Z is representable (cf. VIII 5.1), and its reductive center is the unit subgroup.*
- (c) *Suppose that the reductive rank of G is locally constant (cf. 1.7(b)). Then G admits a reductive center Z . If G/Z is representable—for example, if G is affine over S —then the maximal tori T of G (respectively, the Cartan subgroups C of G) and the maximal tori T' of $G' = G/Z$ (respectively, the Cartan subgroups C' of G') are in bijective correspondence:*

$$T' = T/Z, \quad C' = C/Z,$$

and conversely

$$T = \varphi^{-1}(T'), \quad C = \varphi^{-1}(C'),$$

where $\varphi : G \rightarrow G'$ is the canonical homomorphism.

- (d) *Let T be a maximal torus of G , let $\mathfrak{g}$ denote the Lie algebra of G , and consider the homomorphism*

$$\theta : T \longrightarrow \mathrm{GL}(\mathfrak{g})$$

induced by the adjoint representation of G (II 4). Then $\ker \theta$ is a reductive center of G .

Proof. (a) and (b). The proof of (a) and of the first assertion of (b) is essentially identical to that of 1.7(a) and (b), and we shall not repeat the argument. We point out only that in (a) one must appeal to IX 5.6, using the fact that G has connected fibers.

We prove the second assertion of (b). Suppose that Z is a reductive center of G , and put $G' = G/Z$. By VIII 5.1 the quotient is representable. It is affine over S , of finite presentation over S by VIII 5.8, and even smooth over S : this may be checked on the fibers because G' is flat and of finite presentation over S , and VI_B 9.2(xii) states that a quotient of a smooth algebraic group over a field is smooth. Since G has connected fibers, so does G' . Thus G' satisfies the same hypotheses as G .

By 4.3 we may reduce to the case in which S is the spectrum of a field. Let Z' be a central subgroup of multiplicative type in G' ; it remains to prove that Z' is the unit subgroup. Let Z_1 be the inverse image of Z' in G , and consider the action of Z_1 on G by inner automorphisms. Since Z is central, it acts trivially, and hence Z_1 acts through an

action of Z' on G . Moreover Z' acts trivially both on Z and on $G/Z = G'$. Since Z is central, the action of Z' on G therefore comes from a group homomorphism

$$u : Z' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(G', Z),$$

characterized by

$$r(z') \cdot g = g \cdot u(z')(\varphi(g)),$$

where $r : Z' \rightarrow \mathrm{Aut}_{S\text{-gr}}(G)$ is the representation just considered and $\varphi : G \rightarrow G'$ is the canonical homomorphism. Giving u is equivalent to giving a group homomorphism

$$v : G' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(Z', Z).$$

By X 5.8 the group on the right is representable and étale over S . Its unit section is therefore an open immersion, so $\ker v$ is an open subgroup of G' . Since G' has connected fibers, $\ker v = G'$. Thus $v = 0$, hence $u = 0$, and Z' , and therefore Z_1 , acts trivially on G . Consequently Z_1 is central in G . It is a commutative extension of the group Z' of multiplicative type by the group Z of multiplicative type, and hence, over a field, is itself of multiplicative type by 7.1.1. Being central in G , it is contained in the reductive center Z . Thus $Z_1 = Z$, whence Z' is the unit subgroup. This proves (b).

(d) Put $Z = \ker \theta$, a subgroup of multiplicative type in T , for example by IX 6.8. By 4.5, every central homomorphism $u : H \rightarrow G$, with H of multiplicative type and of finite type, factors through T , and therefore through Z . Since formation of Z commutes with every base change, it remains to show that Z is central, or equivalently that its centralizer C is equal to G . By XI 5.3, C is a closed smooth subgroup of G . Since Z acts trivially on $\mathfrak{g}$, we have

$$\mathrm{Lie}(C) = \mathfrak{g}.$$

It follows that $C = G$. Indeed, the immersion $C \rightarrow G$ is étale, as may be checked fiberwise: it is an unramified homomorphism of smooth algebraic groups of the same dimension (VI_B 1.3), so X 3.5 applies. Hence $C \rightarrow G$ is an étale closed immersion, and therefore an open immersion by SGA 1, I 5.1. Since it is both open and closed and G has connected fibers, it is an isomorphism.

(c) By 1.7(b), G admits a maximal torus locally for the fpqc topology. Proposition 4.2 and part (d) therefore show that G admits a reductive center Z . Part (d) also shows that every maximal torus T of G contains Z . Clearly $T' = T/Z$ is a torus. To prove that it is a maximal torus of G' , Definition 1.3 reduces us to the case in which S is the spectrum of an algebraically closed field; the assertion then follows from *BIBLE* 7, Th. 3(a).

Conversely, let T' be a maximal torus of G' , and set $T = \varphi^{-1}(T')$. We prove that T is a maximal torus of G . The question is local for the fpqc topology, so we may suppose that G already admits a maximal torus T_0 . By what precedes, $T'_0 = T_0/Z$ is a maximal torus of G' . By the conjugacy theorem 1.7(c), T' and T'_0 are locally conjugate for the fpqc topology. Since $\varphi : G \rightarrow G'$ is a covering for this topology, sections of G' lift locally to sections of G ; thus T and T_0 are likewise locally conjugate. Hence T is a maximal torus. The argument for Cartan subgroups is identical. This completes the proof.

Here is a useful reformulation of (d) when T is diagonalizable, say $T = D_S(M)$, where M is a finite-type free $\mathbb{Z}$ -module. By I 4.7.3, the T -module $\mathfrak{g}$ decomposes into the direct sum of its T -submodules $\mathfrak{g}_m$, $m \in M$:

$$\mathfrak{g} = \bigoplus_{m \in M} \mathfrak{g}_m.$$

These submodules are necessarily locally free. Suppose that for every $m \in M$ the rank of $\mathfrak{g}_m$ is constant, as is the case in particular when S is connected. Then the set

$$R = \{m \in M \mid \mathfrak{g}_m \neq 0\}$$

of “roots” is finite. We obtain:

Corollary 4.8. *Under the preceding hypotheses and with the preceding notation, the reductive center of G is the intersection of the kernels in T of the root characters $m \in R$. Thus there is an isomorphism*

$$Z \simeq D_S(N),$$

where N is the quotient of M by the subgroup generated by R .

Corollary 4.9. *Let G be an algebraic group over an algebraically closed field k . Suppose that G is smooth, connected, and affine over k , that its reductive center is the unit subgroup, and that its Lie algebra $\mathfrak{g}$ is nilpotent. Then G is “unipotent,” that is, it admits a composition series whose quotients are isomorphic to $\mathbb{G}_a$.*

By BIBLE 6, Th. 4, Cor. 3, it is enough to prove that a maximal torus T of G is the unit subgroup, or equivalently that its Lie algebra $\mathfrak{t}$ is zero. The T -module $\mathfrak{g}$ decomposes according to the characters of T by I 4.7.3, so for every $t \in \mathfrak{t}$, the endomorphism $\text{ad}_{\mathfrak{g}}(t)$ of $\mathfrak{g}$ is semisimple. If $\mathfrak{g}$ is nilpotent, this endomorphism is also nilpotent, hence zero. By 4.7(d), the hypothesis on the reductive center makes the homomorphism

$$T \longrightarrow \text{GL}(\mathfrak{g})$$

a monomorphism, and it therefore induces an injective map on Lie algebras. By II 4.5, this says that $\text{ad}_{\mathfrak{g}}(t) = 0$ implies $t = 0$. Consequently $\mathfrak{t} = 0$, as required.

Proposition 4.10. *Let G be a smooth connected affine algebraic group over an algebraically closed field k . Then the reductive center of G is the intersection of the maximal tori of G .*

The intersection is meant schematically, or equivalently in the sense of subfunctors of G : it is the largest closed subscheme of G contained in every maximal torus. Since G is noetherian, it is already the intersection of a suitable finite family of maximal tori.

Let Z denote this intersection. It is a closed subgroup of a torus, hence is of multiplicative type, and by 4.5 it contains the reductive center of G . To prove equality it remains to show that Z is central. Since G is connected, IX 5.5 reduces this to showing that Z is invariant. By construction Z is invariant under every inner automorphism $\text{int}(g)$, with $g \in G(k)$. Its normalizer N , represented by VIII 6.7, is therefore a closed subgroup of G containing all rational points of G . Since G is reduced, $N = G$, which proves the result.

Proposition 4.11. *Let S be a prescheme and let G be a smooth affine S -group prescheme with connected fibers and unipotent rank zero. Then G has locally constant reductive rank by 1.9, and hence the “prescheme of maximal tori of G ,” denoted by $\mathcal{T}$, is defined by 1.10 and is smooth, separated, and of finite type over S . Let G act on $\mathcal{T}$ through inner automorphisms, giving a homomorphism of group functors*

$$u : G \longrightarrow \text{Aut}_S(\mathcal{T}).$$

Under these hypotheses the following three subfunctors of G are identical:

- (i) the reductive center Z of G ;
- (ii) the center Z' of G ;
- (iii) the kernel $Z'' = \ker u$ of the preceding homomorphism.

In particular, the center of G is represented by a subgroup of multiplicative type in G .

Proof. Clearly $Z \subset Z' \subset Z''$. It remains to prove $Z'' \subset Z$, or, since the hypotheses are stable under base change, that every section g of G over S which acts trivially on $\mathcal{T}$ is a section of Z . Put $G' = G/Z$. By 4.7(b) and (c), the prescheme $\mathcal{T}'$ of maximal tori of G' is canonically isomorphic to $\mathcal{T}$. We are therefore reduced to the case $G = G'$, in which the reductive center Z is the unit subgroup. Notice that 4.7(c) also shows that the unipotent rank of G' equals that of G , and hence is zero.

We must prove that g is the unit section of G . The usual reduction process reduces us first to the case in which S is noetherian, and then to the case in which S is local artinian: when S is locally noetherian, it is enough to check the assertion after every base change $S' \rightarrow S$ with S' local artinian. In this case $Z'' = \ker u$ is representable by VIII 6.2(a) and 6.5(c); alternatively, representability follows from XI 6.8. By Nakayama's lemma it is enough to show that the fiber Z''_0 is the unit subgroup. We may therefore suppose that S is the spectrum of a field k , which we may take to be algebraically closed.

The subgroup Z'' lies in the stabilizer of every point of $\mathcal{T}(k)$, that is, in the normalizer $N(T)$ of every maximal torus T of G . Since the unipotent rank of G is zero, XI 5.9 says that T is an open subgroup of $N(T)$. Thus

$$\mathrm{Lie} N(T) = \mathrm{Lie} T,$$

and $\mathrm{Lie} Z''$ is contained in $\mathrm{Lie} T$ for every maximal torus T . Proposition 4.10 shows that the intersection of the Lie algebras of all maximal tori is the Lie algebra of the reductive center Z , which is zero in the present reduction. Hence $\mathrm{Lie} Z'' = 0$, so Z'' is étale over k .

Moreover, Z'' is invariant in G . Since G is connected, it follows that Z'' is central in G . Thus it is contained in

$$T = \mathrm{Centr}_G(T)$$

for every maximal torus T , and therefore in their intersection. By 4.10 this intersection is $Z = 0$, completing the proof.

5 Application to the prescheme of subgroups of multiplicative type⁶

Theorem 5.1. *Let S be a prescheme, let G be a smooth affine S -group prescheme, and let M be the “prescheme of subgroups of multiplicative type in G ,” representing the functor described in XI 4.1. For every integer $n > 0$, let T_n be the subfunctor of M for which $T_n(S')$ is the set of subgroup preschemes H of multiplicative type in $G_{S'}$ satisfying*

$$n \mathrm{id}_H = 0.$$

Thus T_n is representable and affine over S by XI 3.12(a). Let

$$u_n : M \longrightarrow T_n$$

be the morphism defined by $u_n(H) = {}_nH$, where ${}_nH = \ker(n \mathrm{id}_H)$. Then:

⁶For a generalization of the results of this section to the case in which G is not assumed affine over S , see M. Raynaud, “Ample sheaves on group schemes and homogeneous spaces,” Chapter IX.

- (a) Every subprescheme U of M of finite type over S is contained in a closed subprescheme of finite type over S . Every closed subprescheme X of M of finite type over S is affine over S .
- (b) Suppose that S is quasi-compact, and let X be a closed subprescheme of M of finite type over S . Then there is an integer $n > 0$ such that, for every multiple m of n , the induced morphism

$$u_m|_X : X \longrightarrow T_m$$

is a closed immersion.

Proof. (a) For the first assertion, take for X the schematic closure of U (EGA I, 9.5.1 and 9.5.3). This closure is defined because the immersion $i : U \rightarrow M$ is quasi-compact: U is of finite type over S , and M is separated over S by XI 4.1. It remains to prove that such an X is affine over S , which will prove the second assertion at the same time. The question is local on S , so we may suppose that S is affine. Since U is of finite type over S , it is contained in a quasi-compact open of M . We are therefore reduced to the case in which U itself is an open subprescheme of finite type over S , hence quasi-compact. Notice that a closed subscheme of a scheme affine over S is affine over S .

It is enough to show that such a U is contained in a closed affine subprescheme of M . In this form the usual reduction process reduces us immediately to the case in which S is noetherian. The same argument applies to (b), reducing it to the case in which S is affine noetherian.

Recall the inverse limit T of the T_n used in the proof of XI 4.1. It is an affine prescheme over S , though not of finite type, and its local rings are noetherian, as was observed at the beginning of the proof of IX 3.7. More explicitly, if $t = (t_n) \in T$, then the transition morphisms $T_m \rightarrow T_n$ are smooth and the dimensions of the T_m are bounded. Hence there is an n_0 such that $T_n \rightarrow T_{n_0}$ is étale at t_n for every multiple n of n_0 . The proof just cited, or XI 3.11, shows that the canonical morphism

$$u : M \longrightarrow T$$

is an immersion and induces isomorphisms on local rings. It need not be an open immersion or a quasi-compact morphism.

Let U be a quasi-compact open of M . We shall show that its closure in T is contained in M . If X is the schematic closure of U in M , this will make

$$u|_X : X \longrightarrow T$$

a closed immersion. Then X is affine, proving (a). Since U is noetherian it has only finitely many irreducible components. Every $t \in T$ in the closure of U is a specialization of a point $x \in U$. The local ring of t in T is noetherian, and so is the local ring A of t in $\overline{\{x\}}$, endowed with the induced reduced structure.

The canonical morphism from the noetherian integral local scheme

$$S' = \text{Spec}(A)$$

to T sends its closed point to t and its generic point to $x \in M$. We must show that $t \in M$. Replacing A , if necessary, by the quotient by a minimal prime of a suitable complete local A -algebra flat over A (EGA 0_{III}, 10.3.1), we may suppose that A is complete and has algebraically closed residue field. One could even reduce further to a discrete valuation ring by EGA II, 7.1.7. Renaming S' as S , we are reduced to the following lemma.

Lemma 5.2. *Let S be the spectrum of a complete noetherian integral local ring with algebraically closed residue field. Let G be a smooth affine S -group prescheme, let $(H(n))_{n>0}$ be a system of subgroups of multiplicative type in G , indexed by the positive integers, and let η be the generic point of S . Suppose that:*

(a) *if m is a multiple of n , then*

$$H(n) = {}_nH(m);$$

(b) *there is a subgroup H_η of multiplicative type in G_η such that*

$$H(n)_\eta = {}_n(H_\eta)$$

for every $n > 0$.

Then there is a subgroup H of multiplicative type in G such that $H(n) = {}_nH$ for every $n > 0$.

Proof. For every n , let

$$C_n = \text{Centr}_G(H(n)).$$

This is a closed subgroup prescheme of G , smooth over S , by XI 5.3. The positive integers being ordered by divisibility, the C_n form a decreasing family of closed subpreschemes of G . Since G is noetherian, this family is stationary. Let C be the common value of the C_n for sufficiently divisible n . Then C satisfies the same conditions as G , and all the $H(n)$ are in fact subgroups of C . Replacing G by C , we may therefore suppose that all the $H(n)$ are central subgroups of G .

Let s be the closed point of S , and let Z_s be the reductive center of G_s , defined in 4.4. By the definition in 4.1, Z_s contains every $H(n)_s$. Since A is complete, Z_s lifts to a subgroup prescheme Z of multiplicative type in G , by XI 5.8. We claim that Z contains every $H(n)$. Indeed, $H(n)$ is central and hence commutes with Z in G . The resulting homomorphism

$$Z \times H(n) \longrightarrow G$$

has, by IX 6.8, an image K which is a subgroup of multiplicative type in G containing Z . It remains only to prove $K = Z$; this follows from $K_s = Z_s$ and IX 5.1 bis.

We are thus reduced to proving the analogue of the lemma with G replaced by a group Z of multiplicative type and of finite type over S , not necessarily smooth over S . Since A is complete with algebraically closed residue field, Z is diagonalizable by X 3.3 and 1.4, hence is of the form $D_S(M)$, where M is a finitely generated commutative group. Every subgroup of multiplicative type H of Z is therefore diagonalizable by IX 2.11(i), and so is of the form $D_S(N)$, where N is a quotient group of M , by VIII 3.2. Thus the $H(n)$ correspond to quotient groups $M(n)$ of M , and the existence of a subgroup H such that $H(n) = {}_nH$ for every n is equivalent to the existence of a quotient group N of M such that

$$M(n) = N/nN$$

for every n . This follows at once from the existence of the subgroup H_η of G_η such that $H(n)_\eta = {}_n(H_\eta)$ for every n . This proves Lemma 5.2 and hence Theorem 5.1(a).

(b) We already know that

$$u|_X : X \longrightarrow T$$

is a closed immersion. It follows readily that, for all sufficiently large m , the composite

$$u_m|_X : X \longrightarrow T_m$$

is also a closed immersion. Since we shall not need this fact below, we omit the details of the proof.

Corollary 5.3. *With the notation of 5.1, let U be a subprescheme of M , both open and closed, and of finite type over S . Then U , with the structure induced by M , is affine over S . If S is quasi-compact, there is an integer $n > 0$ such that, for every multiple m of n , the induced morphism*

$$u_m|_U : U \longrightarrow T_m$$

is both an open and a closed immersion; equivalently, it is an isomorphism onto an open-and-closed subprescheme of T_m , endowed with the induced structure.

Proof. The first assertion follows from 5.1(a), and the second from 5.1(b), since $u_m : M \rightarrow T_m$ is smooth by XI 2.2 bis and a smooth immersion, that is, an étale immersion, is an open immersion.

Corollary 5.4. *Let S be a prescheme, let G be a smooth affine S -group prescheme whose reductive rank is locally constant (1.7(b)), and let $\mathcal{T}$ be the “prescheme of maximal tori of G ” (1.10). Then $\mathcal{T}$ is smooth and affine over S . If T is a maximal torus of G and $N(T)$ its normalizer, then $G/N(T)$ (XI 5.3 bis) is affine over S . The same is true of $G/C(T)$, where $C(T)$ is the centralizer of T , provided that*

$$W(T) = N(T)/C(T)$$

is finite over S ; cf. 2.1(b).

Proof. The second assertion is contained in the first, since the conjugacy theorem identifies $G/N(T)$ with $\mathcal{T}$, by XI 5.5 bis. For the first assertion, observe that by construction $\mathcal{T}$ is isomorphic to an open-and-closed subprescheme of M . Indeed, one may suppose that the reductive rank of G is constant and equal to r ; then $\mathcal{T}$ is the subprescheme of M corresponding to the subtori of relative dimension r , that is, the largest subprescheme of M over which the universal subgroup of multiplicative type

$$H \subset G_M$$

is a torus of relative dimension r . Corollary 5.3 therefore applies. Finally, $G/C(T)$ is finite over

$$G/N(T) \simeq (G/C(T))/W(T),$$

and hence is affine because $G/N(T)$ is affine.

Corollary 5.5. *Let G be a smooth affine algebraic group over a field k . Then the scheme M of subgroups of multiplicative type in G is a disjoint union of affine k -schemes. For every subgroup H of multiplicative type in G , if C and N denote respectively its centralizer and its normalizer, then the quotients G/C and G/N are affine.*

Proof. By XI 5.1 bis, the saturation under the action of G on M of every finite closed subset of M is open. Indeed, one reduces to the case in which k is algebraically closed, and hence to the orbit of a k -rational point x ; then, by the cited result, the morphism

$$G \longrightarrow M, \quad g \longmapsto g \cdot x$$

is smooth and therefore open, so its image is open.

Let U be the union of the equivalence classes of the closed points of M under the action of G . Then U is open and contains every closed point of M ; by the Nullstellensatz, $U = M$. Thus M is a disjoint union of open subpreschemes, which are necessarily also closed. Hence M is the sum prescheme of subpreschemes M_i , each of which is the G -orbit

of a closed point. Each M_i is quasi-compact, hence of finite type, and is therefore affine by 5.3.

If H is a subgroup of multiplicative type in G , it corresponds to a k -rational point x of M , and G/N identifies with the orbit of x under G , by XI 5.5 bis. Thus G/N is affine. Since C is an open subgroup of N , by XI 5.9, the quotient G/C is finite over G/N ; indeed,

$$G/N \simeq (G/C)/(N/C).$$

Consequently G/C is affine as well.

This proof also gives the following result.

Corollary 5.6. *Under the hypotheses of 5.5 on G and H , the subprescheme U of M consisting of the subgroups of multiplicative type in G which are locally conjugate to H is an open-and-closed subprescheme of M .*

In picturesque language, every subgroup H' of multiplicative type in G which is a limit of subgroups conjugate to H is itself conjugate to H .

Remarks 5.7. Let S be a prescheme, let G be a smooth affine S -group prescheme, let H be an S -prescheme of multiplicative type and of finite type, and put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G).$$

By XI 4.2, M is representable, smooth over S , and separated over S . One can prove for M a result entirely analogous to 5.1, either by reducing to 5.1 through an argument analogous to that of XI 4.2, or directly by an argument modeled on the proof of 5.1. The corresponding variants of 5.3, 5.5, and 5.6 follow; we leave their formulation to the reader.

6 Maximal tori and Cartan subgroups of not necessarily affine algebraic groups (algebraically closed base field)

Lemma 6.1. *Let G be a connected algebraic group over a field k , and let Z be an algebraic subgroup of finite index in the center of G . Then G/Z is affine.*

Proof. We may suppose that Z is the center of G , since a scheme finite over an affine scheme is affine. Consider the vector spaces

$$P^n = P^n(G) = \mathcal{O}_{G,e}/\mathfrak{m}_{G,e}^{n+1} \quad (n \geq 0),$$

where $\mathfrak{m}_{G,e}$ is the maximal ideal at the identity. The group G acts on the $P^n(G)$ through the adjoint representation. Let Z_n be the kernel of the corresponding homomorphism

$$\mathrm{ad}_n : G \longrightarrow \mathrm{GL}(P^n).$$

Using the connectedness of G , one readily verifies that Z is the intersection of the Z_n . Since G is noetherian, this intersection is equal to one of the Z_n . Passing to the quotient, ad_n defines a monomorphism

$$G/Z_n \longrightarrow \mathrm{GL}(P^n),$$

which is a closed immersion. Hence G/Z_n is affine, and so is G/Z .

Lemma 6.2. *Let G be a smooth algebraic group over a field k , let Z be a central algebraic subgroup, put $G' = G/Z$, and let*

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Let T be a subgroup of multiplicative type in G , let $T' = u(T)$, let C be the centralizer of T in G , and let C' be the centralizer of T' in G' . Then

$$C'^0 \subset u(C).$$

Proof. We may suppose that k is algebraically closed. Put

$$C_1 = (u^{-1}(C'^0))_{\text{red}}.$$

It is enough to prove that $C_1 \subset C$. Indeed, $u(C_1)$ is connected and of finite index in C'^0 , hence is equal to it set-theoretically and therefore scheme-theoretically, since C' , and consequently C'^0 , is smooth.

The commutator morphism

$$(g, t) \longmapsto gtg^{-1}t^{-1}$$

from $G \times G$ to G induces a morphism

$$\varphi : C_1 \times T \longrightarrow Z_1, \quad Z_1 = Z_{\text{red}}.$$

Indeed, the source is reduced, and for $g \in C_1(k)$ and $t \in T(k)$, the element $gtg^{-1}t^{-1}$ belongs to $Z(k)$ because g centralizes T modulo Z .

A calculation on k -rational points shows that $\varphi(g, t)$ is additive in g and in t , hence “bilinear.” We claim that this homomorphism is identically zero; this proves $C_1 \subset C$. By the density theorem for T , we reduce to the case in which T is finite, so that $n \text{id}_T = 0$ for some integer $n > 0$. The morphism φ is then defined by a group homomorphism

$$C_1 \longrightarrow \text{Hom}_{k\text{-gr}}(T, Z_1).$$

Since Z_1 is commutative and smooth, the group on the right is represented by an étale algebraic group over k . Since C_1 is connected, every group homomorphism from C_1 to that group is zero.

Corollary 6.3. *Under the preceding hypotheses, suppose that C' is connected. Then*

$$C' = u(C), \quad C = u^{-1}(C').$$

Proof. In this case $C' = C'^0$. Moreover, C plainly contains Z , and hence

$$C = u^{-1}(u(C)).$$

Lemma 6.4. *Let G be an algebraic group over a field k , and let Z be a central algebraic subgroup such that*

$$G/Z = G'$$

is a torus. Then G is commutative. If k is algebraically closed, there is a torus T in G such that $u(T) = G'$, where

$$u : G \longrightarrow G/Z = G'$$

is the canonical homomorphism.

Proof. We may suppose that k is algebraically closed. Consider again the commutator morphism

$$G \times G \longrightarrow G, \quad (g, h) \longmapsto ghg^{-1}h^{-1}.$$

Since Z is central, this morphism factors through a morphism $G' \times G' \rightarrow G$. Since G' is commutative, the latter takes its values in Z , and even in Z_{red} , because $G' \times G'$ is reduced. As above, the resulting morphism

$$\varphi : G' \times G' \longrightarrow Z_{\text{red}}$$

is bilinear and hence zero. Thus G is commutative.

To find a torus T in G with $u(T) = G'$, we may, after replacing G by G_{red}^0 , suppose that G is smooth and connected. Replacing Z by Z_{red}^0 , we may also suppose that Z is smooth and connected. By a well-known theorem of Chevalley,⁷ Z is an extension of an abelian variety by an affine group. We are therefore reduced to the two cases in which Z is affine or Z is an abelian variety. In the first case G is affine and the result is classical (BIBLE 7, Theorem 3(a)).

Suppose then that Z is an abelian variety. Every homomorphism from the additive group $\mathbb{G}_a$ to the torus $G' = R$, or to the abelian variety Z , is trivial. The same is therefore true of every homomorphism from $\mathbb{G}_a$ to G ; thus G contains no subgroup isomorphic to $\mathbb{G}_a$. Chevalley's structure theorem gives an exact sequence

$$(*) \quad 0 \longrightarrow T \longrightarrow G \longrightarrow A \longrightarrow 0,$$

where A is an abelian variety and T is a smooth connected affine group. Since T is commutative and contains no additive subgroup, T is a torus; it is plainly the unique maximal torus of G . It remains to prove that every epimorphism

$$u : G \longrightarrow R$$

to a torus R satisfies $u(T) = R$.

Put

$$\text{Hom}_{\text{gr}}(T, \mathbb{G}_m) = M, \quad \text{Hom}_{\text{gr}}(R, \mathbb{G}_m) = P.$$

These are finitely generated free $\mathbb{Z}$ -modules, and

$$T = D_k(M), \quad R = D_k(P).$$

Let

$$B = \text{Ext}_{k\text{-gr}}^1(A, \mathbb{G}_m);$$

this is also the set of k -rational points of the abelian variety dual to A . We have

$$(**) \quad \text{Ext}^1(A, T) = \text{Hom}_{\text{gr}}(M, B), \quad \text{Ext}^1(A, R) = \text{Hom}_{\text{gr}}(P, B).$$

In particular, the extension G of A by T is determined by a homomorphism

$$\theta : M \longrightarrow B.$$

The exact sequence $(*)$ also gives

$$0 \longrightarrow \text{Hom}(A, R) \longrightarrow \text{Hom}(G, R) \longrightarrow \text{Hom}(T, R) \longrightarrow \text{Ext}^1(A, R).$$

⁷Editor's note: references should be supplied here.

Moreover, $\text{Hom}(A, R) = 0$, and the last map identifies with the homomorphism

$$\text{Hom}(P, M) \longrightarrow \text{Hom}(P, B)$$

induced by $\theta : M \rightarrow B$. Setting

$$N = \ker \theta,$$

we obtain canonical bijections

$$\text{Hom}(G, R) \simeq \text{Hom}(P, N) \simeq \text{Hom}(S, R), \quad S = D_k(N).$$

This has the following immediate geometric description.

Lemma 6.5. *Let G be an extension of an abelian variety A by a torus $T = D_k(M)$, defined by a homomorphism*

$$\theta : M \longrightarrow B = \text{Ext}^1(A, \mathbb{G}_m),$$

over an algebraically closed field k . Put $N = \ker \theta$, and let $S = D_k(N)$ be the corresponding torus, isomorphic to T/U , where $U = D_k(M/N)$. Consider the extension $H = G/U$ of A by S . This extension splits,⁸ so there is a unique projection from H to S , and hence a unique homomorphism

$$v : G \longrightarrow S$$

extending the canonical homomorphism $T \rightarrow S$. For every torus R and every homomorphism $u : G \rightarrow R$, there is a unique homomorphism $u' : S \rightarrow R$ such that $u = u'v$. In particular,

$$\text{Im}(u) = \text{Im}(u') = u(T).$$

It follows in particular that if u is an epimorphism, then its restriction to T is also an epimorphism. This completes the proof of 6.4.

Theorem 6.6. *Let G be a smooth connected algebraic group over an algebraically closed field k .*

- (a) *The maximal tori of G are conjugate.*
- (b) *If T is a torus in G , then its centralizer is smooth and connected.*
- (c) *The assignment*

$$T \longmapsto \text{Centr}_G(T)$$

gives a bijection between the maximal tori T of G and the Cartan subgroups of G defined in No. 1. An algebraic subgroup C of G is a Cartan subgroup if and only if it is smooth, nilpotent, and set-theoretically equal to its connected normalizer; in that case it is even scheme-theoretically equal to its connected normalizer. Moreover,

$$C = \text{Centr}_G(T),$$

where T is the unique maximal torus of C , and

$$\text{Norm}_G(C) = \text{Norm}_G(T).$$

⁸Editor's note: the extension is split.

- (d) Let $v : G \rightarrow H$ be an epimorphism of smooth connected algebraic groups. The maximal tori, respectively the Cartan subgroups, of H are the images of the maximal tori, respectively the Cartan subgroups, of G . If T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$.
- (e) Under the hypotheses of (d), suppose that $\ker v$ is a central subgroup of G . Then

$$T \mapsto v(T), \quad C \mapsto v(C)$$

give bijections between the maximal tori, respectively the Cartan subgroups, of G and those of H . The Cartan subgroups of G contain the center of G , and hence $\ker v$; they are precisely the groups $v^{-1}(C')$, where C' is a Cartan subgroup of H .

We also record the following immediate consequence of Theorem 6.6(c).

Corollary 6.7. *The group G is nilpotent, that is, $G(k)$ is nilpotent, if and only if every torus in G is central, or equivalently if G has a unique maximal torus. In that case this torus is the largest subtorus of G .*

Proof of Theorem 6.6. Let Z be a central algebraic subgroup of G , put $G' = G/Z$, and let

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Then G' is smooth and connected. If T' is a subtorus of G' , it follows from 6.4 that $u^{-1}(T')$ is commutative and that T' is the image of a subtorus of $u^{-1}(T')$, hence of a subtorus of G . Since $u^{-1}(T')$ is commutative, it plainly has a largest subtorus T : the sum of two subtori is a third subtorus containing both. Thus

$$u(T) = T'.$$

It follows immediately that, for every maximal torus T of G , its image $T' = u(T)$ is a maximal torus of G' , and that

$$T \mapsto u(T)$$

is a bijection between the maximal tori of G and those of G' .

Now take

$$Z = \text{Centr}(G)_{\text{red}}^0.$$

Then G' is affine by 6.1. Its maximal tori are conjugate, and therefore so are those of G , which proves (a). Moreover, G is nilpotent, respectively has a unique maximal torus, if and only if G' has the same property. Since G' is affine, these two conditions on G' are equivalent (BIBLE 6, Th. 4, Cor. 2); hence they are also equivalent for G . If G has a unique maximal torus T , then T is invariant and therefore central; and because every torus of G is contained in a maximal torus, every torus is central. Conversely, if every torus is central, then so are the maximal tori, and the conjugacy theorem (a) shows that there is only one. This proves 6.7.

Let T be any torus of G , let $T' = u(T)$, and put

$$C' = \text{Centr}_{G'}(T').$$

The group C' is connected (BIBLE 6, Th. 6(a)). By 6.3 the centralizer C of T is $u^{-1}(C')$, and is therefore connected because Z is connected. This proves (b). If T is maximal, then T' is maximal; hence C' is nilpotent, and so is C , which is a central extension of C' .

Moreover, T is a maximal torus of C , and by 6.7 it is the unique maximal torus of C . Consequently

$$T \longmapsto \text{Centr}_G(T)$$

is a bijection from the maximal tori of G to the Cartan subgroups of G . We also have

$$\text{Centr}_G(T) = C \subset \text{Norm}_G(C) \subset \text{Norm}_G(T).$$

The centralizer of T has finite index in its normalizer (cf. XI 5.9); the argument there remains valid without the affineness hypothesis, because it uses only the representability of the two functors in question, as noted in XI 6.5. Since C is smooth and connected, it follows that

$$C = \text{Norm}_G(C)^0.$$

Furthermore, the bijection between maximal tori and Cartan subgroups shows that the normalizers of C and T have the same k -valued points. Since the latter normalizer is smooth,

$$\text{Norm}_G(T) = \text{Norm}_G(C).$$

To finish the proof of (c), it remains to show that a smooth, connected, nilpotent subgroup C of G which has finite index in its normalizer is a Cartan subgroup. Since Z is central, the normalizer of C contains Z ; and since Z is smooth and connected, it follows that $Z \subset C$. Hence

$$C = u^{-1}(C'), \quad C' = u(C),$$

and

$$\text{Norm}_G(C) = u^{-1}(\text{Norm}_{G'}(C')).$$

Thus C' is connected and nilpotent and has finite index in its normalizer. By Theorem 1 of BIBLE 7, it is a Cartan subgroup of G' , and it follows at once that C is a Cartan subgroup of G .

We next prove (e). We are in the situation considered at the beginning of the proof, with $\ker u = Z$ and $G' = H$, and we have already seen that $T \mapsto u(T)$ is a bijection between the maximal tori of G and those of G' . Using the bijection just proved between maximal tori and Cartan subgroups, we obtain a bijection between Cartan subgroups C of G and Cartan subgroups C' of G' : to

$$C = \text{Centr}_G(T)$$

we associate

$$C' = \text{Centr}_{G'}(T'), \quad T' = u(T).$$

Since C' is connected by (b), 6.3 gives

$$C' = u(C), \quad C = u^{-1}(C'),$$

which proves (e).

It remains to prove (d). Let T be a maximal torus of G . We first prove that $v(T)$ is a maximal torus of H ; by the conjugacy theorem (a), this will imply that all maximal tori of H arise in this way. Let R be a torus of H containing $v(T)$. Replacing H by R and G by $v^{-1}(R)_{\text{red}}^0$, we may suppose that $R = H$, so that H is a torus; it remains to prove $v(T) = H$. Put again

$$Z = \text{Centr}(G)_{\text{red}}^0, \quad G' = G/Z, \quad H' = H/v(Z).$$

We have a commutative diagram with exact rows

$$(D) \quad \begin{array}{ccccccc} e & \longrightarrow & Z & \longrightarrow & G & \xrightarrow{u} & G' \longrightarrow e \\ & & \downarrow v'' & & \downarrow v & & \downarrow v' \\ e & \longrightarrow & v(Z) & \longrightarrow & H & \xrightarrow{u'} & H' \longrightarrow e. \end{array}$$

We already know that $u(T)$ is a maximal torus of G' . Because G' is affine, the epimorphism

$$v' : G' \longrightarrow H'$$

is an epimorphism of smooth connected affine groups. Therefore $v'(u(T))$ is a maximal torus of H' (BIBLE 7, Th. 3(a)), and hence equals H' . Thus $u'(v(T)) = H'$, so in order to prove $v(T) = H$ it is enough to show that $v(T) \supset v(Z)$. Now $v(Z)$ is a smooth connected subgroup of H , hence a torus, and

$$v'' : Z \longrightarrow v(Z)$$

is an epimorphism. By 6.4,

$$v(Z) = v''(S)$$

for a torus S of Z . Since S is central in G , it is contained in the maximal torus T , and therefore $v(Z) \subset v(T)$. This proves (d) for maximal tori.

In view of the bijection between maximal tori and Cartan subgroups, it remains to prove that, if T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$. Return to diagram (D), now without assuming that H is a torus, and put

$$T' = u(T), \quad C' = u(C).$$

By (e), C' is the centralizer of the maximal torus T' . Since G' , and hence H' , is affine, $v'(C')$ is the centralizer of the maximal torus $v'(T')$ of H' (BIBLE 7, Th. 3(a)). Equivalently, $u'(v(C))$ is the centralizer of $u'(v(T))$. Because C contains Z , the group $v(C)$ contains $v(Z)$. It is therefore the inverse image under u' of $u'(v(C))$, that is, of the centralizer of $u'(v(T))$. By (e), applied to $u' : H \rightarrow H'$ and the maximal torus $v(T)$ of H , this inverse image is precisely the centralizer of $v(T)$. This completes the proof of 6.6. $\square$

7 Application to smooth group preschemes not necessarily affine

Theorem 7.1. *Let S be a prescheme and let G be an S -group prescheme that is smooth, separated, and of finite type over S . Suppose that G admits a maximal torus locally for the faithfully flat quasi-compact topology. Then:*

(a) *The map*

$$T \longmapsto \text{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups of G . If C corresponds to T , then T is the unique maximal torus of C .

- (b) Let T, T' be two maximal tori of G , and let C, C' be the corresponding Cartan subgroups. Then

$$\mathrm{Transp}_G(T, T') = \mathrm{Transp}_G(C, C') = \mathrm{Transp}_G(T, C').$$

The first two terms are also identical to the strict transporters. Moreover, the functor in question is representable by a closed subscheme of G , smooth over S . The tori T, T' and the Cartan subgroups C, C' are locally conjugate for the étale topology.

- (c) A maximal torus of G , and a Cartan subgroup of G , exist locally for the étale topology.
- (d) Suppose that every finite subset of a fiber G_s is contained in an affine open of G —for example, that G is quasi-projective over S , or that S is artinian. Then the functor

$$\mathcal{T} : (\mathrm{Sch})^\circ / S \longrightarrow (\mathrm{Sets})$$

defined in 1.10, the functor of maximal tori of G , is isomorphic to the functor

$$\mathcal{C} : (\mathrm{Sch})^\circ / S \longrightarrow (\mathrm{Sets})$$

of Cartan subgroups of G , and is representable by a smooth, separated prescheme of finite type over S . This prescheme is quasi-projective over S when G is, and is affine over S when G is affine over S , or when S is artinian.

- (e) Let $u : G \rightarrow G'$ be a morphism of S -group preschemes, where G' is smooth, separated, and of finite type over S , and suppose that

$$u_s(G_s) = G'_s$$

for every $s \in S$, that is, that u_s is faithfully flat.⁹ For every maximal torus T of G , $u(T)$ is a maximal torus of G' . If G' has connected fibers, then for every Cartan subgroup C of G , $u(C)$ is a Cartan subgroup of G' ; and if C is the centralizer of T , then $u(C)$ is the centralizer of $u(T)$. In either case the induced morphism $T \rightarrow u(T)$, respectively $C \rightarrow u(C)$, is faithfully flat.

- (f) Under the hypotheses of (e), suppose in addition that $\ker u$ is a central subgroup of G . Then

$$T \longmapsto u(T)$$

is a bijection from the set of maximal tori of G onto the set of maximal tori of G' . If G' has connected fibers, then

$$C \longmapsto u(C)$$

is likewise a bijection from the set of Cartan subgroups of G onto the set of Cartan subgroups of G' ; if $C' = u(C)$, then $C = u^{-1}(C')$.

⁹The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Proposition II.5.1(c).

Remarks 7.2. We shall see in Exposé XV that the conclusion of Theorem 7.1(d) remains valid without the restrictive condition on the finite subsets of the G_s . We shall also prove there the conclusions concerning Cartan subgroups alone that occur in (b), (c), (d), and (e), under the weaker hypothesis that G admit a Cartan subgroup locally for the faithfully flat quasi-compact topology, but not necessarily a maximal torus. For that purpose we shall moreover need to use 7.1 when S is artinian. The method of Exposé XV also considerably simplifies the proof of (c) and (d) when S is not artinian.

Proof of Theorem 7.1. (a) Proceeding as in 3.2, one sees that for every maximal torus T of G ,

$$C = \text{Centr}_G(T)$$

is indeed a Cartan subgroup of G , and that T is determined by C as the unique maximal torus of C . It remains only to show that every Cartan subgroup C of G is defined by a maximal torus of G , or equivalently that C admits a central maximal torus.

The question is still local for the fpqc topology, so we may suppose that G admits a maximal torus T . By XI 6.2,

$$\text{Transp}_G(T, C)$$

is representable by a closed subscheme S' of G , smooth over S ; moreover $S' \rightarrow S$ is surjective by 6.6(c). After the base change $S' \rightarrow S$, we may therefore suppose that there is a section g of G such that

$$\text{ad}(g) \cdot T \subset C.$$

Then $\text{ad}(g) \cdot T$ is a maximal torus of C . It is central fiber by fiber by 6.6(c), and hence central by IX 5.6(b). This proves (a).

(b) Let T, T' be two maximal tori of G , and C, C' their corresponding Cartan subgroups. The following conditions are equivalent:

$$(1) T \subset T', \quad (2) T = T', \quad (3) T \subset C', \quad (4) C \subset C', \quad (5) C = C'.$$

This follows immediately from (a). Applying the same result after arbitrary base change gives the asserted identities among the transporters and strict transporters.

We have already observed in (a) that $\text{Transp}_G(T, C')$ is representable by a closed subscheme of G , smooth over S , whose structural morphism is surjective. By Hensel's lemma this prescheme has a section locally for the étale topology. Thus T, T' , on the one hand, and C, C' , on the other, are locally conjugate for the étale topology. This proves (b).

(c) Suppose first that S is artinian and local. If G admits a maximal torus T , the conjugacy theorem just proved shows that the functor $\mathcal{T}$ of maximal tori of G is represented by the homogeneous space G/N , where N is the normalizer of T in G ; it is smooth by (b). Moreover, as observed in VI_A 3.2.1, every finite subset of $\mathcal{T}$ is contained in an affine open, because $\mathcal{T}$ is a homogeneous space under G .

In general there is a finite extension k'/k of the residue field of S such that $G_{k'}$ has a maximal torus. The extension k'/k is obtained by a finite flat base change $S' \rightarrow S$, and the maximal torus of $G_{k'}$ lifts to a maximal torus of $G_{S'}$ by XI 2.1 bis. Thus $\mathcal{T}_{S'}$ is represented by a smooth, separated prescheme of finite type over S' , every finite subset of which is contained in an affine open. Its natural descent datum is effective. Hence $\mathcal{T}$ is representable, and descent shows that it is smooth, separated, and of finite type over S . By smoothness and Hensel's lemma, $\mathcal{T}$ has a section locally for the étale topology. This proves (c) and (d) when S is artinian. We shall prove below that in this case $\mathcal{T}$ is in fact affine over S .

Now let S be arbitrary. To prove (c) and (d), which are local assertions on S for the Zariski topology, we may suppose that S is affine, that there is a prime ℓ prime to all residue characteristics of S , and that the reductive rank of the fibers of G is constant, say equal to r . The latter rank is visibly locally constant because of the assumed fpqc-local existence of a maximal torus.

Choose a faithfully flat quasi-compact morphism $S' \rightarrow S$, with S' affine, such that $G_{S'}$ admits a maximal torus T_0 , and let C_0 be its centralizer. I claim that there is a suitable power n of ℓ such that

$$C_0 = \text{Centr}_{G_{S'}}({}_nT_0).$$

Indeed, this reduces immediately to the case where S' is noetherian, which was treated in XI 6.2. Fix such an n , and put

$$M = (\mathbb{Z}/n\mathbb{Z})^r, \quad P = \text{Hom}_{S\text{-gr}}(M_S, G).$$

Then P is represented by a closed subprescheme of finite presentation of $({}_nG)^r$, where

$${}_nG = \text{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S, G).$$

This is also the kernel of the n -th-power morphism on G , and is therefore represented by a closed subprescheme of finite presentation of G . Moreover, P is smooth over S by XI 2.1.

Let $s \in S$. From what precedes, there is a finite separable extension $k''/\kappa(s)$ for which $G_{k''}$ has a maximal torus T'' . Since ${}_nT''$ is étale over k'' , we may, after replacing k'' by a finite separable extension, suppose that

$${}_nT'' \simeq M_{k''}.$$

Choose an étale morphism $S'' \rightarrow S$ and a point $s'' \in S''$ above s , inducing the residue-field extension $\kappa(s'') \simeq k''$. We obtain a section of $P_{S''} \otimes \kappa(s'')$ over $\text{Spec } \kappa(s'')$. By Hensel's lemma, after replacing the pointed prescheme (S'', s'') by an étale neighborhood if necessary, this section extends to a section of $P_{S''}$. Equivalently, there is a homomorphism

$$M_{S''} \longrightarrow G_{S''}$$

inducing the given isomorphism

$$M_{\kappa(s'')} \simeq {}_nT''.$$

By IX 6.4—which here reduces to Nakayama's lemma—this homomorphism is a closed immersion above an open neighborhood of s'' , which we may take to be all of S'' . Let H'' be its image, and let C'' be its centralizer in $G_{S''}$; by XI 6.2, C'' is a closed smooth subprescheme of $G_{S''}$.

Lemma 7.3. *Under the preceding hypotheses on ℓ, n, r , for every prescheme S'' over S and every maximal torus T'' of $G_{S''}$, one has*

$$\text{Centr}_{G_{S''}}(T'') = \text{Centr}_{G_{S''}}({}_nT'').$$

Indeed, after faithfully flat descent from $S' \times_S S''$, we reduce to the case in which $G_{S''}$ admits a maximal torus T''_1 for which the displayed equality holds. By (b), T'' and T''_1 are locally conjugate for the étale topology, so the same equality holds for T'' .

Apply this result over $\text{Spec } \kappa(s'')$. It shows that the fiber $C''_{s''}$ is a Cartan subgroup of $G_{s''}$. Now, since G is smooth over S , the union of the identity components of the fibers G_s is an open subprescheme of G ; this is the general EGA IV result for smooth morphisms

recorded by the editors without a precise reference. This open is visibly stable under the group law and hence is a subgroup, denoted G^0 . It satisfies the same preliminary hypotheses as G , and maximal tori of G correspond bijectively to maximal tori of G^0 . Thus, for the proof of (c) and (d), we may and do suppose that G has connected fibers. Its Cartan subgroups then also have connected fibers by 6.6(a).

I claim that C'^0 is a Cartan subgroup of $G_{S''}$ above an open neighborhood of s'' . This will complete the proof of (c). The claim follows from the next lemma.

Lemma 7.4. *Under the hypotheses of 7.1, suppose that G has connected fibers. Let C be a closed group subscheme of G , smooth over S , and let $s \in S$ be such that C_s is a Cartan subgroup of G_s . Then C^0 is a Cartan subgroup of G over an open neighborhood of s .*

We reduce easily to the case where S is local and s is its closed point, and to proving there that C^0 is a Cartan subgroup of G . By flat descent we may further suppose that G admits a maximal torus T . By XI 6.2,

$$\mathrm{Transp}_G(T, C)$$

is represented by a closed subscheme of G , smooth over S . The fiber of this transporter at s is nonempty by the hypothesis on C_s and the conjugacy theorem 6.6(a). Faithfully flat descent therefore reduces us to the case where the transporter has a section over S , hence to the case where C contains a maximal torus T of G . Then T is central in C^0 by IX 5.6(a), so

$$C^0 \subset \mathrm{Centr}_G(T).$$

This is an inclusion of smooth group subschemes over S , with connected fibers and the same relative dimension—namely, the dimension of their common fiber at s . It is therefore an equality, proving the lemma and completing (c).

(d) Retain the preceding notation and hypotheses on ℓ, n, r , and the assumption that G has connected fibers. Let

$$Q : (\mathrm{Sch})^\circ / S \longrightarrow (\mathrm{Sets})$$

be the functor defined by

$$Q(S') = \left\{ \begin{array}{l} \text{subgroups of multiplicative type in } G_{S'} \\ \text{of type } M = (\mathbb{Z}/n\mathbb{Z})^r \end{array} \right\}$$

in the sense of IX 1.4. The rule

$$T \longmapsto {}_n T$$

defines a morphism

$$\varphi : \mathcal{T} \longrightarrow Q,$$

which is a monomorphism by Lemma 7.3.

I claim that Q is representable by a separated prescheme of finite presentation over S . As noted in the proof of XI 3.12(a), there is an isomorphism

$$Q \simeq P_0 / \Gamma,$$

where P_0 is the open and closed subscheme of

$$P = \mathrm{Hom}_{S\text{-gr}}(M_S, G)$$

corresponding to monomorphisms $M_{S'} \rightarrow G_{S'}$, as in IX 6.8, and

$$\Gamma = (\mathrm{Aut}_{\mathrm{gr}}(M))_S.$$

The hypothesis that every finite subset of a fiber G_s lies in an affine open is stable under passage to a closed subprescheme and under fiber products. It is therefore inherited by $({}_nG)^r$, then by P , and then by P_0 . Hence Q is represented by a prescheme of finite presentation over S .¹⁰ The same argument shows that Q is quasi-projective, respectively affine, over S when G is. In every case Q is separated over S .

Lemma 7.5. *The preceding morphism $\varphi : \mathcal{T} \rightarrow Q$ is representable by an open immersion.*

In other words, let H be a subgroup of multiplicative type in G , of type M . We must prove that there is an open subset $U \subset S$ with the following property: for every $S' \rightarrow S$, the group $H_{S'}$ is of the form ${}_nT_{S'}$, for a suitable maximal torus $T_{S'}$ of $G_{S'}$, if and only if $S' \rightarrow S$ factors through U .

We may suppose that the nilpotent rank of the fibers of G is constant, say equal to r' ; it is locally constant by (b). Put

$$C = \mathrm{Centr}_G(H),$$

a closed group subprescheme smooth over S by XI 6.2. Replacing S by the open and closed subset where C has relative dimension r' , we may suppose that this is its relative dimension everywhere. Then $H = {}_nT$ for a maximal torus T of G if and only if C^0 admits a central maximal torus T , everywhere of relative dimension r , and $H = {}_nT$. This gives another description of the subfunctor U .

By flat descent, we may suppose that G admits a maximal torus T_1 . Consider the subfunctor

$$R = \mathrm{Transp}_G(T_1, \mathrm{Centr}_{C^0}(C^0)) \subset \mathrm{Transp}_G(T_1, C).$$

The transporter on the right is represented by a smooth prescheme over S by XI 6.2. The one on the left is represented by the induced open subprescheme, as follows immediately from IX 5.6(a); in particular it too is smooth over S . Its structural morphism is therefore open. Replacing S by its image, equipped with the induced structure, we may suppose that this morphism is surjective. Applying 1.13 to C^0 , we find that C^0 admits a central maximal torus T , because it admits one fpqc-locally. Its relative dimension is that of T_1 , namely r . The remaining condition on H is $H = {}_nT$, and by IX 2.10 this again cuts out a suitable open and closed subset of S . This proves the lemma.

Lemma 7.5 shows that $\mathcal{T}$ is represented by a separated prescheme locally of finite presentation over S , and in fact of finite presentation. For the latter assertion, one may either inspect the proof of Lemma 7.5 to see that φ is a quasi-compact open immersion, or descend fpqc-locally to the case where G has a maximal torus T , when

$$\mathcal{T} \simeq G/\mathrm{Norm}_G(T).$$

This expression, or XI 2.1, also shows that $\mathcal{T}$ is smooth over S . If G is quasi-projective over S , then so are Q and $\mathcal{T}$. If G is affine over S , the affineness of $\mathcal{T}$ over S was proved in 5.4. It is unknown here whether, without the hypothesis that G/S is affine, n can be chosen so that the open immersion of Lemma 7.5 is also closed.

¹⁰The editors left the printed reference to Exposé V unspecified.

It remains to show that $\mathcal{T}$ is affine over S when S is artinian. By EGA I 6.1.7, we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. In view of (f), proved below, it suffices to prove the assertion for

$$G/\mathrm{Centr}(G).$$

This quotient is affine by 6.1, so the preceding affine case applies. This proves (d).

(e) By IX 6.8, there is a subtorus T' of G' such that u induces a faithfully flat morphism

$$T \longrightarrow T'.$$

This characterizes T' as the subsheaf $u(T)$ of G' . Let C and C' be the centralizers of T and T' , respectively. We prove that $C \rightarrow C'$ is flat, and faithfully flat when G' has connected fibers. Since C and C' are flat and of finite presentation over S , SGA 1, I 5.9 reduces us to a base field, which we may suppose algebraically closed. Replacing G, G' by their identity components does not change C^0, C'^0 , so we may suppose G, G' connected. The assertion then follows from 6.6(d), together with (a).

(f) By (a) and (e), it is enough to prove the assertion about Cartan subgroups. A Cartan subgroup C of G , being the centralizer of a maximal torus, contains the center of G , and therefore contains $\ker u$. Thus

$$C = u^{-1}(C'), \quad C' = u(C),$$

where C' is the Cartan subgroup of G' supplied by (e). Hence $C \mapsto u(C)$ is injective. To prove surjectivity, it suffices to show that $u^{-1}(C')$ is a Cartan subgroup of G for every Cartan subgroup C' of G' .

The question is fpqc-local. We may suppose that G admits a Cartan subgroup C_1 . Then $C'_1 = u(C_1)$ is a Cartan subgroup of G' , and is fpqc-locally conjugate to C' by (b). Since

$$u^{-1}(C'_1) = C_1$$

is a Cartan subgroup of G , the same is true of $u^{-1}(C')$. This proves (f) and completes the proof of the theorem. $\square$

Alternatively, to prove that $u^{-1}(C')$ is a Cartan subgroup, observe that it is flat over S , since u is; SGA 1, I 5.9 then reduces the assertion, by definition, to the case of a base field, where 6.6(e) applies.

Corollary 7.6. *Under the hypotheses of 7.1(e), G' admits a maximal torus locally for the étale topology, and therefore satisfies the preliminary hypotheses imposed on G . If moreover $\ker u$ is central, then the functors $\mathcal{T}_G, \mathcal{C}_G$ of maximal tori and Cartan subgroups of G are isomorphic to the analogous functors $\mathcal{T}_{G'}, \mathcal{C}_{G'}$ for G' . Consequently, whenever these functors are representable, they are represented by isomorphic S -preschemes.*

Remark 7.7. (a) In contrast with the case where G is affine over S —in fact it is enough that G have affine fibers, as will be seen in Exposé XVI—local constancy of the reductive rank of G does not imply that G admits a maximal torus fpqc-locally. For example, let S be the spectrum of a discrete valuation ring, and let G_1, G_2 be smooth separated group schemes of finite type over S , such that the generic fiber of G_1 is an elliptic curve and its special fiber is $\mathbb{G}_m$, while the generic fiber of G_2 is $\mathbb{G}_m$ and its special fiber is $\mathbb{G}_a$. Put

$$G = G_1 \times_S G_2.$$

Both fibers of G have reductive rank 1, but G plainly does not admit a maximal torus fpqc-locally.

It is, on the other hand, very plausible that the following condition on a smooth separated group of finite type over a prescheme S is sufficient for the étale-local existence of a maximal torus: the reductive rank and the abelian rank of the fibers of G are locally constant functions.

(b) In the proof of 7.1, particularly in (a), we used XI 6.2 in cases where S was not assumed locally noetherian. In the instances at hand, however, the reduction to affine noetherian S is immediate.

Here is a variant of 7.1(b).

Proposition 7.8. *Let G be a smooth S -group prescheme of finite presentation with connected fibers, let H be a smooth S -group prescheme, and let*

$$i : H \longrightarrow G$$

be a monomorphism of S -groups, making H an S -subgroup of G . For every Cartan subgroup C of G ,¹¹ the transporter $\text{Transp}_G(C, H)$ is represented by a closed subprescheme of G , smooth over S .

The representability of the transporter by a closed subprescheme of G , of finite presentation, follows from XI 6.11, because C has connected fibers when G does, by 6.6(b). To prove smoothness, the standard procedure reduces first to affine noetherian S , then to artinian local S , and finally, by descent, to an algebraically closed residue field. In that case C has a maximal torus T , which is also a maximal torus of G . Unless the reductive and nilpotent ranks of the fiber H_0 equal those of G_0 , the transporter is empty. Under this equality one sees, using the connectedness of C and the smoothness of the centralizer in H of a maximal torus of H , that

$$\text{Transp}_G(C, H) = \text{Transp}_G(T, H).$$

The right-hand side is smooth by XI 2.5, and therefore so is the left-hand side.

The same argument proves the more general assertion 7.9(b) below.

Proposition 7.9. *Let G and $i : H \rightarrow G$ be as in 7.8. Suppose moreover that, for every $s \in S$, H_s is connected and has the same reductive rank and the same nilpotent rank as G_s ; equivalently, H_s contains a Cartan subgroup of G_s . Then:*

- (a) *$\text{Norm}_G(H)$ is represented by a closed subprescheme of G , of finite presentation over S , and the canonical monomorphism*

$$H \longrightarrow \text{Norm}_G(H)$$

is an open immersion. Consequently i is an immersion and

$$H = \text{Norm}_G(H)^0.$$

- (b) *For every Cartan subgroup C of G , $\text{Transp}_G(C, H)$ is a closed subprescheme of G , smooth over S , with surjective structural morphism. If C is the centralizer of a maximal torus T of G , then moreover*

$$\text{Transp}_G(T, H) = \text{Transp}_G(C, H).$$

¹¹The proof only requires C to be a smooth subgroup of G whose geometric fibers are connected centralizers of subgroups of maximal tori.

- (c) Let C be a group subscheme of H . Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G .
- (d) If G admits a Cartan subgroup, respectively a maximal torus, locally for the étale topology or for the fpqc topology, then so does H .

Proof. (a) The representability of $\text{Norm}_G(H)$ by a closed subscheme of G , of finite presentation over S , is contained in XI 6.11. Since H is smooth, hence flat and locally of finite presentation over S , VI_B 2.6 reduces the assertion that

$$H \longrightarrow \text{Norm}_G(H)$$

is an open immersion to the fibers. We therefore reduce to the case where $S = \text{Spec } k$, with k algebraically closed.

The assertion then reduces to showing that the corresponding map on Lie algebras is an isomorphism,¹² or equivalently that

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where $\mathfrak{g}, \mathfrak{h}$ are the Lie algebras of G, H and the superscript denotes invariants; compare II 5.2.3(i). By hypothesis H contains a Cartan subgroup C of G , the centralizer of a maximal torus T of G . It therefore suffices to prove

$$(\mathfrak{g}/\mathfrak{h})^T = 0.$$

By complete reducibility of representations of T (I 4.7.3), this follows from

$$(\mathfrak{g}/\mathfrak{c})^T = 0, \quad \mathfrak{c} = \text{Lie}(C).$$

The latter equality is equivalent to

$$\mathfrak{g}^T = \mathfrak{c}.$$

It says that the centralizer C and the normalizer of T have the same Lie algebra, and follows from the fact that C is an open subgroup of that normalizer by XI 5.9. This proves (a).

(b) was proved in 7.8.

(c) Since H is a subscheme of G , the assertion reduces immediately to an algebraically closed base field, where it follows directly from the hypothesis on H .

(d) follows from (b), (c), and the Hensel lemma XI 1.10. $\square$

Corollary 7.10. *Let G, H be as in 7.9, and suppose that for every algebraically closed field k over S , every element of $G_k(k)$ normalizing H_k belongs to $H_k(k)$. Then H is a closed subscheme of G and is its own normalizer.*

This follows immediately from 7.9(a). We shall apply it in particular to Borel subgroups, and more generally to parabolic subgroups, of G .

¹²The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Corollary II.5.6.

Corollary 7.11. *Let G, H be as in 7.9. Then H contains every group subscheme Z of G which is central in G , flat, and of finite presentation over S .*

As usual, we reduce first to affine noetherian S , then to artinian S ; the hypotheses of 7.1 then hold. By 7.9(c), we may suppose that H contains a Cartan subgroup C of G , and it remains to show $C \supset Z$. Since S is artinian, the quotient

$$G' = G/Z$$

is represented by a group prescheme of finite type over S ; the canonical morphism $u : G \rightarrow G'$ is faithfully flat and has kernel Z by VI_A 3.2. The group G' is smooth over S . Theorem 7.1(f) therefore shows that $C = u^{-1}(C')$ for a Cartan subgroup C' of G' , and hence that C contains Z .

Corollary 7.12. *Let G, G' be smooth S -group preschemes of finite presentation, and let $u : G \rightarrow G'$ be a faithfully flat group homomorphism, meaning that u_s is faithfully flat for every $s \in S$. Suppose that $\ker u$ is central and that G has connected fibers. Then*

$$H' \mapsto H = u^{-1}(H')$$

defines a bijection between:

- *the group subschemes H' of G' , smooth and of finite presentation over S , having at every $s \in S$ the same reductive rank and the same nilpotent rank as G' ; and*
- *the group subschemes H of G , smooth and of finite presentation over S , having at every $s \in S$ the same reductive rank and the same nilpotent rank as G .*

Moreover, H has connected fibers if and only if H' does.

Let H be a group subscheme of G with the stated properties. By 7.11 it contains $Z = \ker u$, so faithfully flat descent gives

$$H = u^{-1}(H')$$

for a uniquely determined group subscheme H' of G' . Using 6.6(e), one checks immediately that H' has the asserted properties. If H has connected fibers, so does $H' = H/Z$.

Conversely, start with $H' \subset G'$ having the stated properties, and put $H = u^{-1}(H')$. In view of 6.6(e), it remains to show that H is smooth over S , and that it has connected fibers when H' does. Since u is flat and H' is flat over S , H is already flat over S . It therefore suffices to check smoothness, respectively connectedness, on geometric fibers, reducing to an algebraically closed base field.

Then H' contains a Cartan subgroup C' of G' , and H contains its inverse image C . By 6.6(e), C is a Cartan subgroup of G , hence is smooth and connected. The conclusion now follows from the next lemma.

Lemma 7.13. *Let G, G' be group preschemes flat and of finite presentation over S , let $u : G \rightarrow G'$ be a faithfully flat group homomorphism, and let C' be a group subscheme of G' , of finite presentation over S , such that $C = u^{-1}(C')$ is smooth over S , respectively has connected fibers. For every group subscheme H' of G' , of finite presentation over S , containing C' , and smooth over S , respectively with connected fibers, its inverse image $H = u^{-1}(H')$ is smooth over S , respectively has connected fibers.*

As noted above, this reduces at once to the case where S is the spectrum of a field. The group H is then a principal C -bundle over H/C , by VI_B 9, while

$$H/C \simeq H'/C'$$

by Exposé IV. Since H' is smooth, respectively connected, so is H'/C' , hence so is H/C . The same property holds for C by hypothesis and therefore for the bundle H . This proves the lemma and Corollary 7.12.

8 Semisimple elements; unions and intersections of maximal tori in group schemes not necessarily affine

Throughout this section, G denotes an S -group prescheme that is smooth and of finite presentation over S , with connected fibers.

Suppose first that S is the spectrum of an algebraically closed field k . When G is affine, the notion of a semisimple element of $G(k)$ was defined in *BIBLE* 4, No. 4; one checks at once that this notion is invariant under an algebraically closed extension k'/k of the base field. Moreover, by *BIBLE* 6, Theorem 5(c), an element $g \in G(k)$ is semisimple if and only if it is contained in a maximal torus of G .

When G is no longer assumed affine, Chevalley's theorem gives a canonical expression of G as an extension of an abelian variety by a smooth connected affine algebraic group:

$$(*) \quad e \longrightarrow V \longrightarrow G \longrightarrow A \longrightarrow e.$$

We shall say that $g \in G(k)$ is *semisimple* if it is a semisimple element of $V(k)$. Since the maximal tori of V are precisely the maximal tori of G , this is equivalent to saying that g belongs to a maximal torus of G . This notion is again invariant under algebraically closed extensions of the base field.

Suppose next that $S = \text{Spec } k$ for an arbitrary field k , and let $g \in G$. After choosing an algebraically closed extension K of $\kappa(g)$, the point g is the image of a geometric point g' of G with values in an algebraically closed extension K of k . We call g semisimple when g' is semisimple. What precedes shows that this is independent of the choice of g' . If k'/k is any extension, the set of semisimple elements of $G_{k'}$ is the inverse image of the set G^{ss} of semisimple elements of G .

Finally, for arbitrary S , a point $g \in G$ is called semisimple when it is semisimple in its fiber G_s . Thus, for every morphism $S' \rightarrow S$, the subset $G_{S'}^{\text{ss}}$ is the inverse image of G^{ss} . Suppose that the functor $\mathcal{T}$ of maximal tori defined in 1.10 is representable by a prescheme of finite presentation over S . This is so, for example, if G admits a maximal torus locally for the fpqc topology, by 7.1(d), at least when G is quasi-projective over S . Let T be the canonical maximal torus of $G_{\mathcal{T}}$, the “universal maximal torus of G ”, and consider the morphism

$$u : T \longrightarrow G$$

induced by the projection $G_{\mathcal{T}} \rightarrow G$. It follows immediately from the definition that G^{ss} is the image of this morphism. We obtain:

Proposition 8.1. *The set G^{ss} of semisimple elements of G is locally constructible. It is therefore constructible when S , and hence G , is quasi-compact and quasi-separated.*

As usual, we reduce to affine noetherian S . By the standard noetherian criterion for constructibility (EGA 0_{III}, 9.2.3), we may further suppose that S is integral and need

only show that there is a nonempty open $U \subset S$ such that $G^{\text{ss}}|_U$ is constructible. After shrinking U , and replacing it if necessary by a finite covering, we may suppose that G is separated over S and contains a maximal torus. By 7.1(d), $\mathcal{T}$ is then represented by a prescheme of finite presentation over S ; the same is true of its universal torus, whose image in G is therefore constructible.

Suppose again that k is a field, and let H be the subgroup of G generated by the preceding morphism (VI_B 1.2). It is a smooth group subscheme of G , connected because the universal torus is connected, and its formation commutes with every extension of the base field, as does the formation of $\mathcal{T}$.

When k is algebraically closed, H is also the algebraic subgroup of G generated by the maximal tori of G , or, equivalently, by all the tori of G . It is likewise the smallest algebraic subgroup of G containing all semisimple elements of $G(k)$. These characterizations remain valid whenever k is infinite, because, as will be proved in Exposé XIV, the set of k -rational points of $\mathcal{T}$ is dense in $\mathcal{T}$.

The subgroup H is invariant in G . Indeed, it is enough to check this after passing to an algebraically closed field; since H and G are then smooth, it suffices to check stability under the inner automorphisms $\text{int}(g)$, $g \in G(k)$, which is immediate. One can even show that H is characteristic in G , that is, stable under $\text{Aut}_{k\text{-gr}}(G)$.

It follows from 6.6(d) that G/H has reductive rank zero. More precisely, if K is an invariant algebraic subgroup of G , then, over an algebraically closed field, the maximal tori of G/K are the direct images of the maximal tori of G . Hence G/K has reductive rank zero if and only if K contains all maximal tori of G , or, over an arbitrary field, if and only if $K \supset H$. Thus H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero. It is also the smallest algebraic subgroup of G having the same reductive rank as G .

Finally, H is affine. This may again be checked after passing to an algebraic closure. In the exact sequence (*), every maximal torus of G is contained in V ; consequently $H \subset V$, and V is affine. We have proved:

Proposition 8.2. *Let G be a smooth connected algebraic group over a field k , and let $\bar{k}$ be an algebraic closure of k . There exists an algebraic subgroup H of G such that $H_{\bar{k}}$ is the algebraic subgroup of $G_{\bar{k}}$ generated by the maximal tori, or equivalently by all the tori, of $G_{\bar{k}}$. The group H is also characterized as the smallest algebraic subgroup of G having the same reductive rank as G , or as the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero. It is a smooth, connected, invariant, affine subgroup of G , and its formation commutes with every extension of the base field.*

The characterization in terms of G/H gives the following consequences.

Corollary 8.3. *Let G be a smooth connected algebraic group over an algebraically closed field k . Then G has reductive rank zero —equivalently, in the notation of 8.2, $H = (e)$ — if and only if G is an extension of an abelian variety by a smooth connected unipotent algebraic group, that is, a successive extension of groups isomorphic to the additive group $\mathbb{G}_a$.*

Indeed, Chevalley's exact sequence (*) reduces the assertion to the fact that a smooth connected affine group V has reductive rank zero if and only if it is unipotent; this is contained in BIBLE 6.4, Theorem 4, Corollary 3. Thus, in the notation of 8.2, H is the smallest invariant algebraic subgroup of G such that $(G/H)_{\bar{k}}$ is an extension of an abelian variety by a smooth connected affine unipotent algebraic group.

Corollary 8.4. *In the notation of 8.2, $G = H$, or equivalently $G_{\bar{k}}$ is generated by its maximal tori, if and only if G is affine and every homomorphism $G_{\bar{k}} \rightarrow \mathbb{G}_a$ is trivial.*

Remarks 8.5.

- (a) Let V be the largest smooth connected affine algebraic subgroup of $G_{\bar{k}}$, so that $G_{\bar{k}}$ is an extension of an abelian variety by V . It is well known, by Rosenlicht,¹³ that when k is not perfect, V is not in general defined over k : there need not exist an algebraic subgroup V_0 of G such that $(V_0)_{\bar{k}} = V$. If, however, V is generated by its maximal tori—equivalently, if V has no quotient isomorphic to $\mathbb{G}_{a,\bar{k}}$ —then such a group does exist, namely the group H of 8.2. Thus, in this rationality question as in many others (compare Exposé XIV, No. 6), all the difficulties arise from unipotent groups, hence from the additive group, whereas the presence of sufficiently many groups of multiplicative type makes descent work.
- (b) In the notation of 8.2, let H' be the inverse image in G of the commutator subgroup of G/H . Then H' is the smallest invariant algebraic subgroup of G such that $(G/H')_{\bar{k}}$ is a commutative extension of an abelian variety by a smooth connected affine unipotent algebraic group. It is again a smooth connected affine subgroup of G .

Assume that the characteristic p is positive, and let H'' be the inverse image in G of $p^n(G/H')$ for n sufficiently large. Then H''/H' is an abelian variety, and H'' is the smallest invariant algebraic subgroup of G such that G/H'' is a commutative smooth connected affine unipotent algebraic group. It follows that $V = H_{\bar{k}}$, in the notation of (a), if and only if $H'' = G$, or equivalently if and only if every homomorphism $G_{\bar{k}} \rightarrow \mathbb{G}_a$ is trivial.

We now generalize to smooth groups with connected fibers the notion of reductive center developed in Section 4, following the idea of 4.10. Let $\mathcal{Z}$ be the subfunctor of G defined by

$$\mathcal{Z}(S') = \{ g' \in G_{S'}(S') : \text{for every } S'' \rightarrow S' \text{ and every maximal torus } T'' \subset G_{S''}, \\ \text{the pullback } g'' \text{ of } g' \text{ is a section of } T'' \text{ over } S'' \}.$$

Let $\mathcal{T}(S')$ be the set of maximal tori of $G_{S'}$, and let $\tilde{\mathcal{T}}(S')$ be the set of pairs (T', g') where T' is a maximal torus of $G_{S'}$ and g' is a section of T' over S' . Then $\tilde{\mathcal{T}}$ is a subfunctor of

$$\mathcal{T} \times_S G = \mathcal{T}_G,$$

and in this notation one may also write

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \tilde{\mathcal{T}}/\mathcal{T}_G.$$

Using XI 6.8 and 7.1(d), which under suitable hypotheses represents $\mathcal{T}$ by a smooth S -prescheme, one could deduce representability of $\mathcal{Z}$ under corresponding hypotheses. We shall instead obtain it more directly below.

¹³The editors ask that references be supplied here.

Definition 8.6. *Let G be an S -group prescheme, smooth and of finite presentation with connected fibers. We say that G admits a reductive center if the preceding functor $\mathcal{Z}$, which is evidently a subgroup of G , is representable by a group of multiplicative type. In that case $\mathcal{Z}$ is called the reductive center of G .*

If Z is a reductive center of G , then $Z_{S'}$ is a reductive center of $G_{S'}$ for every base change $S' \rightarrow S$. The existence of a reductive center is consequently local for the fpqc topology. The terminology is justified because $\mathcal{Z}$ is invariant under

$$\mathrm{Aut}_S(G),$$

and hence is an invariant subgroup of G ; by IX 5.5, it is central.

Lemma 4.5 is replaced here by the following statement.

Lemma 8.7. *Let $u : H \rightarrow G$ be a central homomorphism, where H is of multiplicative type and of finite type over S . Suppose that for every algebraically closed field k over S , the subgroup $u_k(H_k)$ is contained in the largest smooth connected affine subgroup of G_k . Then u factors through every maximal torus T of G . In particular, u factors through the subfunctor $\mathcal{Z}$ defined above.*

An easy variant of IX 5.1 bis, with equality replaced by inclusion, reduces the assertion to the case in which S is the spectrum of an algebraically closed field: first reduce to affine noetherian S , then to artinian S , and use IX 3.6. Since T is contained in the largest smooth connected affine subgroup V of G , we are reduced to $G = V$, hence to the affine case proved in 4.5.

Proposition 8.8. *Let G be an S -group prescheme, smooth and of finite presentation with connected fibers.*

- (a) *If $S = \mathrm{Spec} k$, then G admits a reductive center Z . When G is an extension of an abelian variety by a smooth connected affine algebraic group V —for example, when k is algebraically closed—then Z is also the reductive center of V , and it is the largest central subgroup of multiplicative type of V .*
- (b) *Let Z be a group subprescheme of multiplicative type in G . Then Z is a reductive center of G if and only if Z_s is a reductive center of G_s for every $s \in S$. It is then the largest subgroup of multiplicative type K in G such that K_s is contained in the reductive center of G_s for every $s \in S$. More generally, let $u : H \rightarrow G$ be a homomorphism with H of multiplicative type and of finite type over S , and suppose that for every algebraically closed field k over S , u_k factors through the largest smooth connected affine subgroup of G_k . Then u factors through Z , and in particular is central.*
- (c) *If G admits a maximal torus locally for the fpqc topology, then G admits a reductive center.*
- (d) *If T is a maximal torus of G , then*

$$T \cap \mathrm{Centr}(G) = \ker(T \rightarrow \mathrm{GL}(\mathfrak{g})),$$

and this group is a reductive center of G .

- (e) Let Z be a reductive center of G , and suppose that $G' = G/Z$ is representable, for example when S is artinian. Then the unit subgroup is a reductive center of G' , and

$$T' \mapsto T = u^{-1}(T')$$

is a bijection from the set of maximal tori of G' to the set of maximal tori of G .

Proof. (a) Let $S = \operatorname{Spec} k$. To prove existence, we may pass to an algebraic closure of k , and thus suppose that G is an extension of an abelian variety by a smooth connected affine algebraic group V . For every S' over k , the maximal tori of $G_{S'}$ are those of $V_{S'}$: this follows from IX 5.2 and from the fact that a homomorphism from a torus to an abelian variety over an algebraically closed field is trivial. Consequently the functor $\mathcal{Z}$ defined using G is the same as the one defined using V , and we reduce to affine G .

The functor $\mathcal{T}$ is representable and is necessarily “essentially free” over k (VIII 6.1); hence $\mathcal{T}_G$ is essentially free over G . Since the universal torus is a closed subscheme of $\mathcal{T}_G = G\mathcal{T}$ (for example by VIII 5.7), VIII 6.4 shows that

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \tilde{\mathcal{T}}/\mathcal{T}_G$$

is represented by a closed subscheme of G . It is therefore a group subscheme. It is of multiplicative type: after passing to an algebraic closure, G admits a maximal torus T , and by definition $Z \subset T$; a subgroup of a group of multiplicative type is of multiplicative type (IX 8). Thus Z is a reductive center. Lemma 8.7 shows that it is the largest central subgroup of multiplicative type of V .

(b) Only sufficiency requires proof. Suppose that Z is of multiplicative type and that Z_s is the reductive center of G_s for every $s \in S$. Lemma 8.7 first shows that Z is contained in every maximal torus of G , and this remains true after every base change. It remains to show that if g is a section of G over S whose pullback to every S' over S is a section of every maximal torus of $G_{S'}$, then g is a section of Z .

As usual, since the fpqc sheaf

$$\mathcal{Z}_0 = \prod_{\mathcal{T}_G/G} \tilde{\mathcal{T}}/\mathcal{T}_G$$

commutes with filtered direct limits of rings, we reduce successively to affine, noetherian, and artinian S . By 7.1(d), $\mathcal{T}$ is then represented by a smooth prescheme over S . Arguing as in (a), $\mathcal{Z}_0$ is represented by a closed subscheme Z_0 of G . We already know $Z \subset Z_0$; by hypothesis $Z_k = (Z_0)_k$ over the residue field k , and Z is flat over S . It follows that $Z = Z_0$, proving that Z is the reductive center. The remaining assertions of (b) follow from 8.7.

(c) follows immediately from (d).

(d) Here $\mathfrak{g}$ denotes the Lie algebra of G , and $T \rightarrow \operatorname{GL}(\mathfrak{g})$ is induced by the adjoint representation. Trivially,

$$T \cap \operatorname{Centr}(G) \subset \ker(T \rightarrow \operatorname{GL}(\mathfrak{g})).$$

The proof of the reverse inclusion is the same as in 4.7(d). Put

$$Z = \ker(T \rightarrow \operatorname{GL}(\mathfrak{g})).$$

Then Z is of multiplicative type (IX 6.8). By (b), we may check that it is a reductive center over a field. Let Z_0 be the reductive center supplied by (a). Plainly

$$Z_0 \subset T \cap \operatorname{Centr}(G) = Z.$$

Conversely, Z is a central subgroup of multiplicative type contained in the smooth connected affine group T , so (a) gives $Z \subset Z_0$. Thus $Z = Z_0$.

(e) Under the hypotheses of 7.1(f), put $Z = \ker u$, and suppose that for every algebraically closed field k over S , Z_k is contained in a maximal torus of G_k . One checks readily that

$$T' \longmapsto u^{-1}(T')$$

is a bijection from the maximal tori of G' to those of G . Applying this observation to the situation of (e) proves the assertion. $\square$

Remarks 8.9.

- (a) The proof of 8.8 is independent of the results of Section 4; in particular, the proof of 8.8(a) does not use 4.4, whose proof is somewhat laborious.
- (b) The subfunctor $\mathcal{Z}$ considered in 8.6 is always central, whether or not it is representable. It may therefore be tempting to call it the reductive center of G in all cases.
- (c) One may likewise generalize 4.9. If G is a smooth connected algebraic group over an algebraically closed field, then G is an extension of an abelian variety by a smooth connected unipotent algebraic group—equivalently, G has reductive rank zero, by 8.3—if and only if the reductive center of G is the unit group and the Lie algebra of G is nilpotent.

9 Complement: actions of group schemes and fixed points

The purpose of this section, added in January 2008, is to study fixed points under the action of a group scheme.

9.1 Representability of the fixed-point functor

Let S be a scheme, let G be an S -group scheme, and suppose that G acts on an S -scheme X . Define the subfunctor X^G of fixed points of X under G by requiring, for every S -scheme T , that $X^G(T)$ be the subset of $X(T)$ consisting of the points fixed under G (VIII 6(e)).

Recall the notion of an S -pure scheme introduced in Raynaud–Gruson, §3.3, and in Raynaud. Let G be a flat S -group scheme of finite presentation. Since the fibers of G have no embedded components, purity takes the following form: G is S -pure if, for every strictly henselian local S -scheme T with closed point t , every generic point $\tilde{x}$ of a fiber of $G \times_S T$ specializes to a point of G_t ; equivalently, the closure of $\tilde{x}$ in $G \times_S T$ meets G_t .

Examples.

- (1) If G is quasi-finite and separated over S , then G is S -pure if and only if it is finite over S .
- (2) If $S = \operatorname{Spec} R$ and $G = \operatorname{Spec} A$, then G is S -pure if and only if A is a projective R -module (Raynaud–Gruson, 3.3.5). In particular, a diagonalizable G is S -pure.
- (3) Every group scheme of multiplicative type is S -pure, since purity is local for the étale topology on S .

- (4) The scheme G is S -pure if it is proper over S , if its fibers are irreducible, or if S is semilocal artinian.
- (5) In particular, every reductive S -group scheme is S -pure.

Proposition 9.2. *Let G be an S -group scheme of finite presentation, flat and S -pure over S , acting on an S -scheme X of finite presentation. Then the fixed-point functor X^G is represented by a closed subscheme of X , of finite presentation over S .*

Indeed, let $a \in X(S)$, and let H be the group subscheme of G fixing a . The point a is fixed under G if and only if $H = G$. The coincidence functor of H and G —whose value on an S -scheme T is a singleton when

$$H \times_S T \longrightarrow G \times_S T$$

is bijective and is empty otherwise—is represented by a closed subscheme of S , defined by a finite-type sheaf of ideals (Raynaud–Gruson, 4.1.1). Apply this result with $S = X$ and with a the universal point of X , that is, the identity element of

$$X(X) = \mathrm{Hom}_S(X, X).$$

9.3 The infinitesimal obstruction

Suppose that G is a flat S -group scheme of finite presentation acting on the left on a smooth S -prescheme X . We shall study the formal smoothness of X^G .

Let S_0 be a closed subscheme of S , defined by a quasi-coherent sheaf of ideals $\mathcal{I}$ with $\mathcal{I}^2 = 0$, and write

$$j : S_0 \longrightarrow S, \quad G_0 = G \times_S S_0, \quad X_0 = X \times_S S_0.$$

Let $\epsilon_0 \in X^G(S_0)$ be a point that lifts to $X(S)$. We shall study the obstruction to lifting ϵ_0 to $X^G(S)$.

Since X is smooth, the set of lifts of ϵ_0 to $X(S)$ is a trivial principal homogeneous space under the abelian group

$$\mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^* \Omega_{X_0/S_0}^1, \mathcal{I})$$

(III 0.2, 0.3). Put

$$L_0 = \mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^* \Omega_{X_0/S_0}^1, \mathcal{I}).$$

This is an $\mathcal{O}_{S_0}$ -module. Since ϵ_0 is fixed under G , it is naturally a G_0 - $\mathcal{O}_{S_0}$ -module. Let

$$\rho_0 : G_0 \longrightarrow \mathrm{Aut}_{S_0}(L_0), \quad \rho : G \longrightarrow \mathrm{Aut}_S(j_* L_0)$$

be the associated representations.

Lemma 9.4. *There is a canonically defined class*

$$c(\epsilon_0) \in H^1(G, j_* L_0) \simeq H^1(G_0, L_0)$$

such that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

The displayed adjunction is the one in III 1.1.2. We work on the small fppf site of S . For every S -scheme T , flat and of finite presentation, write

$$G_T = G \times_S T, \quad X_T = X \times_S T.$$

Consider the sheaf $\mathcal{A}$ on this site defined by

$$\mathcal{A}(T) = \{ \text{lifts of } \epsilon_0 \times_S T \text{ to } X(T) \}.$$

Since the formation of L_0 commutes with flat base change, $\mathcal{A}(T)$ is a trivial principal homogeneous space under

$$H^0(T, j_* L_0).$$

Moreover, because ϵ_0 is fixed, every $g \in G(T)$ acts by affine automorphisms on $\mathcal{A}(T)$, compatibly with the action of G on $j_* L_0$:

$$g(a_T + v) = g(a_T) + \rho(g)(v), \quad v \in H^0(T, j_* L_0).$$

Since G is flat and of finite type, we may take $T = G$ and g equal to the universal point id_G . We thereby obtain an action of the fppf sheaf G on $\mathcal{A}$.

Let a be a lift of ϵ_0 to $X(S)$, and let $g^\sharp$ denote the universal point of G . Define

$$v^\sharp \in H^0(G, j_* L_0)$$

by the relation

$$\rho(g^\sharp)(v^\sharp) = g^\sharp \cdot a_G - a_G.$$

For every S -prescheme Y , set

$$z(Y) : G(Y) \longrightarrow H^0(Y, j_* L_0), \quad g \longmapsto v^\sharp(g).$$

This defines a 1-cocycle $z \in Z^1(G, j_* L_0)$ (Exposé I, §5). Its class $c(\epsilon_0)$ does not depend on the choice of a . In particular, if ϵ_0 lifts to $X^G(S)$, then $c(\epsilon_0) = 0$.

Conversely, if $c(\epsilon_0) = 0$, there is a section

$$w \in H^0(S, j_* L_0)$$

such that, for every S -prescheme Y and every $g \in G(Y)$,

$$z(Y)(g) = g \cdot w_Y - w_Y.$$

Apply this with $Y = G$ and $g = g^\sharp$. It follows that $a - w \in X^G(S)$. Thus ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

Remark 9.5. When S is affine and G is affine, flat, and of finite type over S , this obstruction can be refined to a class

$$\tilde{c}(\epsilon_0) \in H_{G_0}^1(X_0, L_0),$$

where the right-hand side is the G_0 -equivariant cohomology group defined by Wewers (Appendix C). In this case it is unnecessary to assume that ϵ_0 lifts to $X(S)$.

9.6 Smoothness of fixed points

Theorem 9.7. *Let G be a flat S -group scheme of finite type over a noetherian base S , acting on a smooth S -scheme X . Suppose that G is S -pure and that, for every geometric point s of S with algebraically closed residue field,*

$$H^1(G_s, (\Omega_{X_s}^1)^*) = 0.$$

Then the fixed-point functor X^G is represented by a closed subscheme of X , smooth over S .

By 9.2, X^G is represented by a closed subscheme of X , of finite presentation over S . We may suppose $S = \operatorname{Spec} A$ with A local. By the smoothness criterion of SGA 1, III 3.1(iii bis), it is enough to verify formal smoothness for

$$S' = \operatorname{Spec} A'$$

and its closed subscheme

$$S'' = \operatorname{Spec}(A'/I'),$$

where A' is an artinian local ring and I' is an ideal of square zero. By dévissage, we may assume that I' is annihilated by the maximal ideal of A' . If k is the residue field of A' , then I' is therefore a k -vector space.

Let $\epsilon'' \in X^G(S'')$ be a point that we wish to lift to $X^G(S')$, and put

$$\begin{aligned} X' &= X \times_S S', & G' &= G \times_S S', \\ X'' &= X \times_S S'', & G'' &= G \times_S S''. \end{aligned}$$

Since S' is affine and X' is smooth over S' , ϵ'' lifts to $X(S')$. By Lemma 9.4, the obstruction to lifting ϵ'' to $X^G(S')$ is a class

$$c(\epsilon'') \in H^1(G'', L''),$$

where

$$L'' = \operatorname{Hom}_{\mathcal{O}_{S''}}((\epsilon'')^* \Omega_{X''/S''}^1, I').$$

Because L'' is a k -vector space, there is a canonical isomorphism

$$H^1(G'', L'') \simeq H^1(G'' \times_{A'/I'} k, L'').$$

Let $\bar{k}$ be an algebraic closure of k . By IX 3.1 in the affine case, and by Lemma 9.11 in general, there is an isomorphism

$$H^1(G'' \times_{A'/I'} k, L'') \otimes_k \bar{k} \xrightarrow{\sim} H^1(G'' \times_{A'/I'} \bar{k}, L'' \otimes_k \bar{k}).$$

The representation $L'' \otimes_k \bar{k}$ is a direct sum of copies of

$$(\Omega_{X_{\bar{k}}}^1)^*.$$

The group on the right therefore vanishes by hypothesis. It follows that $H^1(G'', L'') = 0$, so the obstruction vanishes and X^G is formally smooth.

Corollary 9.8. *Let G be a flat S -group scheme of finite presentation acting on a smooth S -scheme X of finite presentation. Suppose that G admits a composition series whose factors are of one of the following types:*

- (1) *an abelian S -scheme, that is, a smooth S -group scheme whose fibers are abelian varieties;*
- (2) *an S -group scheme of multiplicative type;*
- (3) *a finite étale S -group scheme of degree invertible on S ;*

(4) a reductive S -group scheme, when S is a $\mathbb{Q}$ -scheme.

Then the fixed-point functor X^G is represented by a closed subscheme of X , of finite presentation and smooth over S .

Indeed, suppose that

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

is an exact sequence of S -groups satisfying the hypotheses. The group G'' acts on $X^{G'}$, and X^G is the fixed-point functor of $X^{G'}$ under G'' . We are therefore reduced to the case of a group of one of the four types listed above. By passage to the limit, we may suppose S noetherian. We verify the cohomological criterion of Theorem 9.7 for a geometric fiber G_s and a finite-dimensional linear representation V of G_s .

For an abelian scheme, every morphism from G_s^n to V is constant; consequently

$$H^i(G_s, V) = 0 \quad (i > 0).$$

For a group scheme of multiplicative type, G_s is diagonalizable, and I 5.3.3 again gives

$$H^i(G_s, V) = 0 \quad (i > 0).$$

In the remaining two cases the assertion follows from semisimplicity of the linear representations of G_s . If G_s is finite étale of degree invertible in $\kappa(s)$, it is a constant finite group of invertible order, and every linear representation of G_s is semisimple. In the reductive case, $\kappa(s)$ is algebraically closed of characteristic zero, and every linear representation of G_s is semisimple (Tauvel–Yu, Theorem 27.3.3).

Remark 9.9. This applies in particular to an action of G/S on a smooth S -group scheme H . If G is of multiplicative type and acts by inner automorphisms on H , with H smooth and affine over S , one recovers XI 5.3: the centralizer of G in H is smooth.

Remark 9.10. When $S = \operatorname{Spec} k$ and G is a linearly reductive algebraic group over an algebraically closed field k , the result is due independently to Fogarty (Theorem 5.4) and Iversen (Proposition 1.3).

The proof of Theorem 9.7 used the following lemma, which is well known for affine group schemes (IX 3.1).

Lemma 9.11. *Let G be a group scheme over an affine scheme $S = \operatorname{Spec} A$, and let $f : S' = \operatorname{Spec} A' \rightarrow S$ be flat. Put $G' = G \times_S S'$. Then, for every quasi-coherent G - $\mathcal{O}_S$ -module M , there are canonical isomorphisms*

$$H^i(G, M) \otimes_A A' \xrightarrow{\sim} H^i(G', M \otimes_A A') \quad (i \geq 0).$$

For $n \geq 1$, the degree- n Hochschild cochains are determined by their value at

$$(\operatorname{id}, \dots, \operatorname{id}) \in G(G) \times \cdots \times G(G).$$

Thus there is an isomorphism of A -modules

$$C^n(G, M) \simeq \Gamma(G^n, M \otimes_A \mathcal{O}_{G^n}).$$

The group $H^n(G, M)$ is the n -th cohomology group of the complex

$$M \longrightarrow \Gamma(G, M \otimes_A \mathcal{O}_G) \longrightarrow \Gamma(G^2, M \otimes_A \mathcal{O}_{G^2}) \longrightarrow \Gamma(G^3, M \otimes_A \mathcal{O}_{G^3}) \longrightarrow \cdots.$$

Since A' is flat over A , EGA IV₁, 1.7.21 gives

$$\Gamma(G^n, M \otimes_A \mathcal{O}_{G^n}) \otimes_A A' \xrightarrow{\sim} \Gamma((G')^n, (M \otimes_A A') \otimes_{A'} \mathcal{O}_{(G')^n}).$$

Taking cohomology proves the assertion.

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